

ENVIRONMENTAL SCIENCE: THEORY INTO PRACTICE-II

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SEMESTER III
COMMON PAPER



DEPARTMENT OF DISTANCE AND CONTINUING EDUCATION
UNIVERSITY OF DELHI



Environmental Science: Theory into Practice–II

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SYLLABUS

Environmental Science: Theory into Practice–II

Syllabus	Mapping
Unit V: Global Environmental Issues and Policies • Causes of Climate change, Global warming, Ozone layer depletion, and Acid rain; Impacts on human communities, biodiversity, global economy, and agriculture • International agreements and programmes: Earth Summit, UNFCCC, Montreal and Kyoto protocols, Convention on Biological Diversity (CBD), Ramsar convention, The Chemical Weapons Convention (CWC), UNEP, CITES, etc. • Sustainable Development Goals: India's National Action Plan on Climate Change and its major missions • Environment legislation in India: Wildlife Protection Act, 1972; Water (Prevention and Control of Pollution) Act, 1974; Forest (Conservation) Act 1980; Air (Prevention & Control of Pollution) Act, 1981; Environment Protection Act, 1986; Scheduled Tribes and other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006.	Lesson 11: Biodiversity and it's Threat (Pages 1-14) Lesson 12: Biodiversity and its Conservation (Pages 15-33) Lesson 13: Sustainable Development & Sustainability (Pages 35-40) Lesson 14: Global Environmental Issues (Pages 41-51)
Unit VI: Biodiversity and Conservation • Definition of Biodiversity; Levels of biological diversity: genetic, species and ecosystem diversity • India as a mega-biodiversity nation; Biogeographic zones of India; Biodiversity hotspots; Endemic and endangered species of India; IUCN Red list criteria and categories • Value of biodiversity: Ecological, economic, social, ethical, aesthetic, and informational values of biodiversity with examples; sacred groves and their importance with examples • Threats to biodiversity: Habitat loss, degradation, and fragmentation; Poaching of wildlife; Man-wildlife conflicts; Biological invasion with emphasis on Indian biodiversity; Current mass extinction crisis • Biodiversity conservation strategies: in-situ and ex-situ methods of conservation; National Parks, Wildlife Sanctuaries, and Biosphere reserves; Keystone, Flagship, Umbrella, and Indicator species; Species reintroduction and translocation • <i>Case studies:</i> Contemporary Indian wildlife and biodiversity issues, movements, and projects (e.g., Project Tiger, Project Elephant, Vulture breeding program, Project Great Indian Bustard, Crocodile conservation project, Silent Valley movement, Save Western Ghats movement, etc)	Lesson 15: Environmental Law & Environmental Legislations in India (Pages 53-64)



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- Human population growth: Impacts on environment, human health, and welfare; Carbon foot-print
- Resettlement and rehabilitation of developmental project affected persons and communities; relevant case studies
- Environmental movements: Chipko movement, Appiko movement, Silent valley movement, Bishnois of Rajasthan, Narmada Bachao Andolan, etc
- Environmental justice: National Green Tribunal and its importance
- Environmental philosophy: Environmental ethics; Role of various religions and cultural practices in environmental conservation
- Environmental communication and public awareness: case studies (e.g., CNG vehicles in Delhi, Swachh Bharat Abhiyan, National Environment Awareness Campaign (NEAC), National Green Corps (NGC) “Eco-club” programme, etc)

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(Pages 65-74)

Lesson 17: Environmental Movements in India
(Pages 75-83)

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LESSON 11

BIODIVERSITY AND IT'S THREAT

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Guest Faculty – Environmental Science
University of Delhi

Structure

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- 11.3 Types of Biodiversity
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11.1 LEARNING OBJECTIVES

After completing the study of this unit you should be able to:

1. List various causes of biodiversity losses and discuss its major impact on the whole world.
2. Explain the need and concept of biodiversity conservation.
3. Enumerate the needs for *in-situ* conservation of species and the various measures and mechanisms.
4. Appreciate the need for *ex-situ* conservation of species and the various mechanisms that are involved in this process of conservation.

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11.2 INTRODUCTION

The word *Biodiversity* refers to the variety of living organisms (flora and fauna). Biodiversity or Biological diversity is defined as the variability among all living organisms from all sources, including terrestrial, marine, and other aquatic ecosystems and ecological complexes of which they are part. *E.O. Wilson* in 1988 defined 'Biological diversity' or 'biodiversity' as that part of nature that includes the differences in genes among the individuals of a species, the variety and richness of all the plant and animal species at different scales in space i.e. local, regional, country wise and global, and various types of ecosystems - both terrestrial and aquatic - within a defined area.

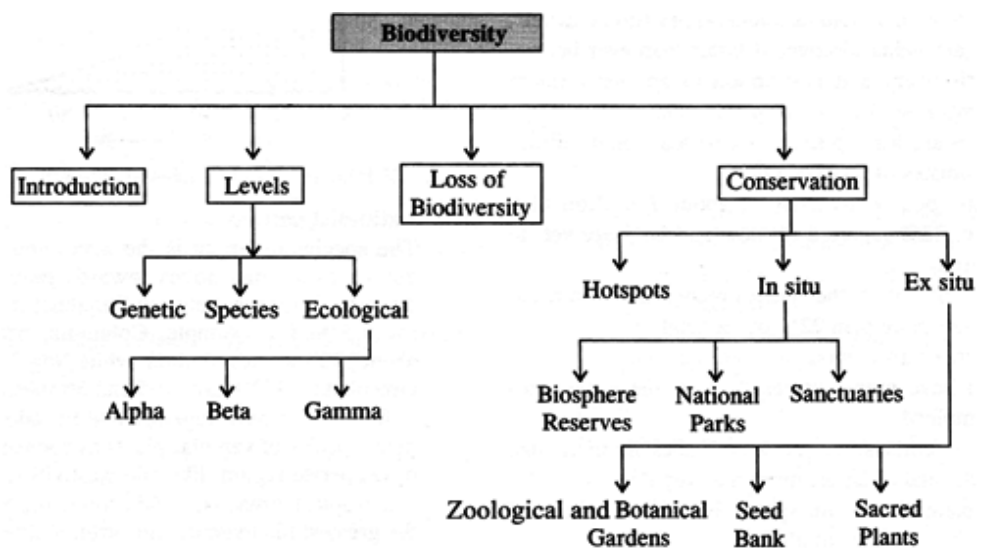


Fig. 1

11.3 TYPES OF BIODIVERSITY

Biological diversity deals with the degree of nature's variety in the biosphere. This variety can be observed at three levels i.e., genetic, species, and ecosystem.



11.3.1 Genetic Diversity

Genetic diversity refers to the variation at the level of individual genes. A tremendous amount of genetic diversity exists within individual species. This genetic variability is responsible for the different characters in species. Genetic diversity is the raw material from which new species arise through evolution. Today, genetic diversity is used to breed new crop varieties and disease-resistant crops.

11.3.2 Species Diversity

The number of species of plants and animals that are present in a region constitutes its species diversity. This diversity is seen both in natural ecosystem and in agricultural ecosystem. Some areas are richer in species than others. For example, natural undisturbed tropical forests have much greater species richness than mono-culture plantations developed by the forest department for timber products. A natural forest ecosystem provides a large number of non-timber forest products that local people depend on such as fruits, fuel, wood, fodder, fiber, gum, resin, and medicines. Timber plantations do not provide the large variety of goods that are essential for local consumption. Modern intensive agroecosystems have a relatively lower density of crops than traditional agro-pastoral farming systems, where multiple crops are planted.

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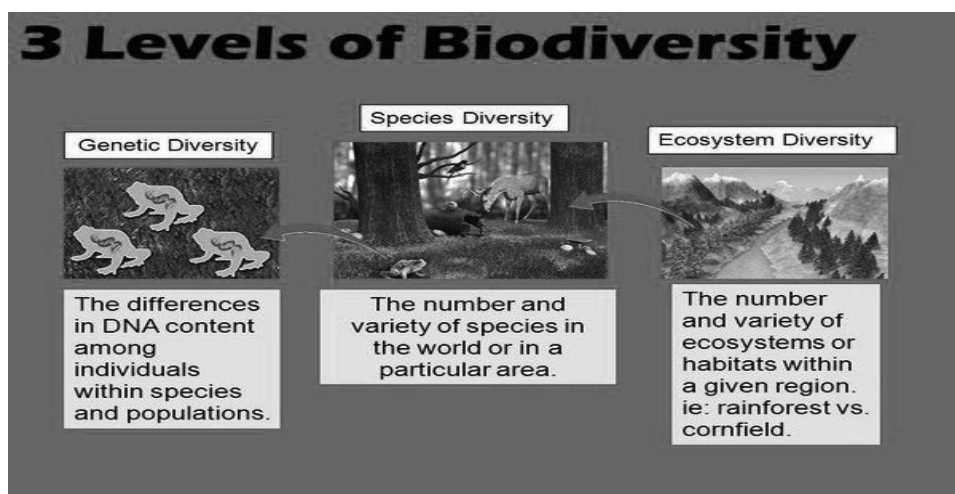


Fig. 2



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Areas that are rich in species diversity are called '*hotspots*' of diversity and the countries with the highest species richness or have a relatively large proportion of these hot spots of diversity are referred to as '*mega diversity nations*'. India is among the world's 15 nations that are exceptionally rich in species diversity. The earth's biodiversity is distributed in specific ecological regions. There are over a thousand major eco-regions in the world. Of these, 200 are said to be the richest, rarest, and most distinctive natural areas. These areas are referred to as the '**Global 200**'. It has been estimated that 50,000 endemic plants which comprise 20% of the global plant life, probably occur in only 25 'hot spots' in the world. These hotspots harbour many rare and endangered species. Two criteria help in defining hotspots namely rich endemism and the degree of threat. To qualify as a hotspot, an area must contain at least 0.5 percent or 1500 of the world's 3,00,000 plant species as endemics (Myers et al., 2000).

11.3.3 Ecosystem Diversity

There are a large variety of different ecosystems on earth, each having its own complement of distinctive interlinked species based on differences in the habitat. Ecosystem diversity can be described for a specific geographical region or a political entity such as a country, a state, or a taluk. Distinctive ecosystems include landscapes like forests, grasslands, deserts, mountains, etc. as well as aquatic ecosystems like rivers, lakes, and seas. Each region also has man-modified areas such as farmland or grazing pastures. It refers to the variation in the structure and functions of the ecosystem. It describes the number of niches, trophic levels, and various ecological processes that sustain energy flow, food webs, and the recycling of nutrients. It has focus on various biotic interactions and the role and functions of keystone species (species determining the ability of a large number of other species to persist in the community), eg. fig, peepal etc.

11.4 METHODS OF MEASURING BIODIVERSITY

There are three perspectives of measuring biodiversity at the level of community. These are (i) Alpha diversity, (ii) beta diversity, and (iii) gamma diversity.



- (i) **Alpha diversity** indicates diversity within the community. It refers to the diversity of organisms sharing the same community or habitat. A combination of species richness and equitability/evenness is used to represent diversity within a community or habitat.
- (ii) **Beta diversity** indicates diversity between communities. Species frequently change when habitat or community changes. There are differences in species composition of communities along with environmental gradients, e.g. altitudinal gradient, moisture gradient, etc. The higher heterogeneity in the habitats of a region or greater dissimilarity between communities exhibit higher beta diversity.
- (iii) **Gamma diversity** refers to the diversity of the habitats over the total land scope or geographical area. The sum of alpha and beta diversities of the ecosystems is an expression of the biodiversity of the landscape, which is considered as Gamma Diversity.

Community diversity refers to the variations in the biological communities in which species live. Higher diversity at the community level provides stability and higher productivity. In temperate grasslands, it has been observed that diverse communities are functionally more productive and stable, even under environmental stresses such as prolonged dry conditions.

11.5 BIOGEOGRAPHIC CLASSIFICATION OF INDIA

Our country can be divided into *ten major regions* based on the geography, climate, and pattern of vegetation seen and the communities of mammals, birds, reptiles, amphibians, insects, and other invertebrates that live in them. Each of these regions contains a variety of ecosystems such as forests, grasslands, lakes, rivers, mountains, and hills which have specific plant and animal species.



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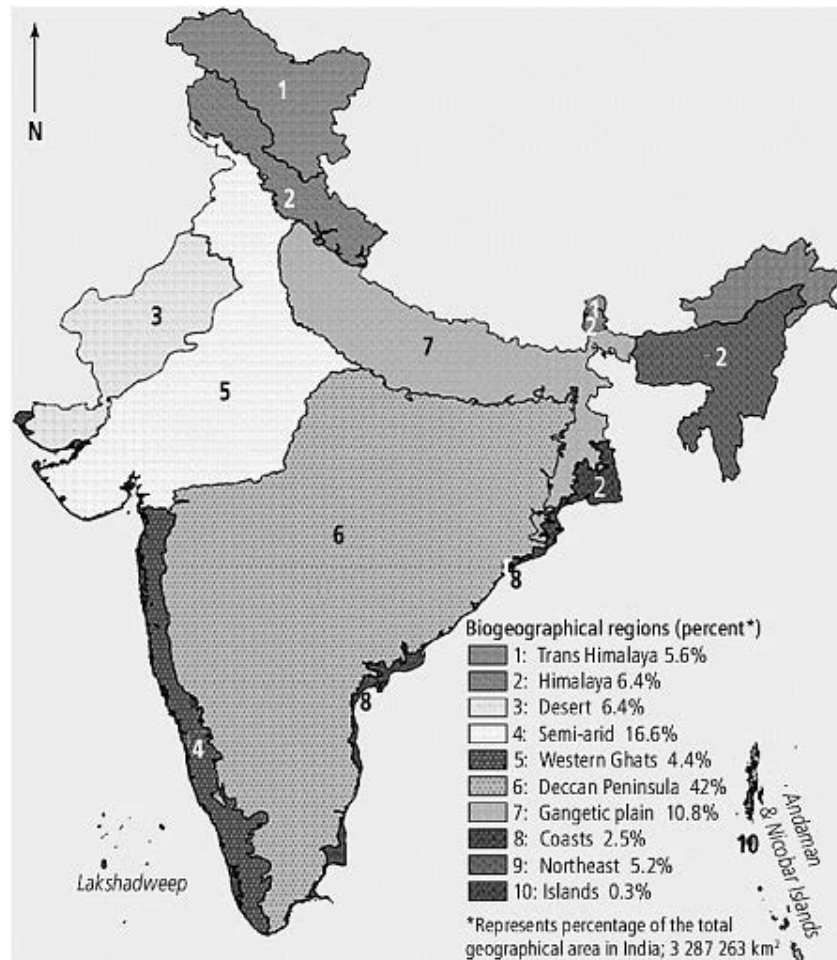


Fig. 3

India's Biogeographic Zones:

1. The cold mountainous snow covered the Trans-Himalayan region of Ladakh.
2. The Himalayan ranges and valleys of Kashmir, Himachal Pradesh, Uttarakhand, Assam, and other north-eastern states.
3. The Terai, the lowland where the Himalayan rivers flow into the plains.



Biodiversity and its Threat

4. The Gangetic and Brahmaputra plains.
5. The Thar Desert of Rajasthan.
6. The semi-arid grassland region of the Deccan plateau, Gujarat, Maharashtra, Andhra Pradesh, Karnataka, and Tamilnadu.
7. The north eastern states of India.
8. The Western Ghats in Maharastra, Karnataka, and Kerala.
9. The Andaman and Nicobar Islands.
10. The long western and eastern coastal belt with sandy beaches, forests, and mangroves.

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11.6 VALUES OF BIODIVERSITY

The value of biodiversity (in terms of its commercial utility, ecological services, social, and aesthetic values) is enormous. There are several ways that biodiversity and its various forms are valuable to humans. The biodiversity value may be classified as follows:

1. Consumptive value:

Biodiversity is an essential requirement for the maintenance of global food supply. The main sources of human food include animals, fish, and plant products. A large number of plants are consumed by human beings as food. A few animal species are consumed by people which comes from cattle, pigs, sheep, goats, buffaloes, chickens, ducks, geese, and turkey species.

Example:

Fish - Many freshwater fish can be grown in ponds. Israel and China already get about half of their fish from aquaculture.

Drugs and medicines - About 75% of the world's population depends upon plants or plant extracts for medicines. The drug Penicillin is used as an antibiotic which is derived from a fungus called *Penicillium*. Likewise, Tetracycline, an antibiotic, obtained from bacteria is used to treat wide variety of infections, Quinine, which is used to cure malaria is obtained from the bark of cinchona tree.

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Fuel - The fossil fuels like coal, petroleum products, and natural gas are the products of biodiversity.

2. Productive value:

Some of the organisms are commercially usable where the product is marketed and sold. The animal products like tusks of elephants, musk from deer, silk from the silkworms, wool from sheep or goats, fur of many animals, etc. all are traded in the market.

Example:

Calabar bean was traditionally used as a poison in West Africa.

Daisy plants were first used as a lice remedy in the Middle East and this led to the discovery of Pyrethrum. Mosquito coils made from Pyrethrum are sold in the market.

The bacterium *Bacillus thuringiensis* produces toxic proteins that kill certain insects.

3. Social value:

These are the values associated with the social life, religion, and spiritual aspects of the people. Many of the plants are considered to be sacred in our country like Tulasi, Mango leaves, Banana leaves, etc. The leaves, fruits, and flowers of some of the plants are used in worship. Many animals like cows, snakes, bulls, and peacocks also have a significant place in spirituality and thus hold special importance. Thus, biodiversity has distinct social values attached to different societies.

4. Ethical value:

The ethical value means that human beings may or may not use a certain species but knowing the very fact that this species exists in nature gives pleasure.

Example:

A peculiar species of Pigeon, a grey/white bird with short legs is no more on this earth.



Similarly, the Dodo species is also no more.

Human beings do not derive anything directly from kangaroos or giraffes but strongly feel that these species should exist in nature.

5. Aesthetic value:

Every one of us would like to visit vast stretches of land to enjoy the visible life. People from farther areas spend a lot of time and money to visit wildlife areas where they can enjoy the aesthetic value of biodiversity and this type of tourism is known as *eco-tourism*.

Eco-tourism is estimated to generate 12 billion dollars of revenue annually which roughly gives the aesthetic value of biodiversity. A study of the impact of environment on the psyche was undertaken by Kaplan and Kaplan (1989) in which they found that being near nature relieved working stresses while people who worked in a closed environment or human-made structures experienced much more job stress and illnesses.

11.7 THREATS TO BIODIVERSITY

Habitat loss is mainly due to human population growth, industrialisation and changes in land use patterns, poaching of wildlife, and man-wildlife conflicts. Man has begun to overuse or misuse most of these natural ecosystems. Due to unsustainable resource use, once productive forests and grasslands have been turned into deserts and wastelands have increased all over the world. Scientists have estimated that human activities are likely to eliminate approximately 10 million species by the year 2050.

1. Human population growth, industrialisation, and changes in the land use patterns:

Around 1.8 million species of plants and animals are known to science. The actual number of species that have been existing is more than 18 million. Though new species have been continually identified, the rate of extinction is very high (10-20,000 species per year i.e., 1000 to 10,000 times faster rate). Human actions are expected to exterminate 25% of the

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world's species in the next 20-30 years. The mega extinction of species is related to human population growth, industrialisation, and changes in the land use patterns in India. The reasons are:

- Forests and grasslands are changed to agricultural land. Encroachments are being repeatedly legalised.
- Natural wetlands are drained to establish crop lands leading to loss of aquatic species.
- Mangroves have been cleared for fuel wood and prawn farming, which has led to a decrease in the habitat essential for the breeding of marine fish.
- Grasslands are changed to other forms, degraded by overgrazing leading loss of cattle, goat, and sheep.
- Natural forests are being deforested for timber and replanted for teak, sal, etc. Such monoculture does not support biodiversity in forests which has closed canopy and rich undergrowth. Excess collection of firewood by lopping off branches of trees' canopy is opened up altering the local biodiversity.
- Foraging cattle retard the regeneration of forest as young seedlings are trampled.
- Ever increasing population gradually decreased the buffer zones and forested areas. A prime example is Gir national park, the last bastion of Asiatic lion with a meter gauge railway line, state expressway, and 3 temples.
- Repeated fires by local grazers to increase growth of grass ultimately reduce regeneration of grasses.
- Introductions of exotic weeds like *Lantana* bushes, *Eupatorium* shrubs, and 'congress' grass are invading at the expense of indigenous undergrowth species. Following traditional farming techniques like slash and burn in Himalayas, and rab, lopping of tree branches for making wood ash fertilizer in Western Ghats are now leading to loss of biodiversity.



- Over harvesting of fish by large trawling boats is leading to depletion of fish stocks. Marine turtles caught in the net are massacred off the coast of Odisha. The rare whale shark, a highly endangered species, is being killed off the coast of Gujarat.

2. Poaching:

Specific threats to certain animals are related to large economic benefits. The skin and bones from tigers, ivory from elephants, horns from rhinos, and perfume from the musk deer are extensively used abroad. Bears are killed for their gall bladders. Corals and shells are also collected for export or sold on the beaches of Chennai, Kanyakumari, and the Andaman and Nicobar islands. Tortoises, exotic birds, and other small animals are packed into tiny containers and smuggled abroad for the pet trade. A variety of wild plants with real or sometimes, dubious medicinal values are being over harvested. The commonly collected plants include *Rauwolfia*, *nux vomica*, *Datura*, etc. The garden plants collected for illegal trade include orchids, ferns, and mosses.

3. Man wild life conflicts:

Conflicting situations with wild life starts causing immense damage and danger to man. Example: In Sambhalpur, Odisha, 195 humans are killed in last 5 years by elephants and in retaliation villagers killed 98 elephants and badly injured more than 30 elephants. Similarly incidents with tigers, leopards, etc. are in news. Shrinking forest cover, human encroachment, ill and weak animals, lack of food (one adult elephant needs 200 kg green fodder and 150 kg of clean water) for animals, protecting villagers by putting electric fence, etc. are the main reasons for such happenings. As the compensation by government is not enough, conflicts occur between forest department and villagers.

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Fig. 4

11.8 ENDANGERED & ENDEMIC SPECIES IN INDIA

The species of plants and animals whose existence is in danger by human activities are called *endangered species*. These endangered species have been categorized into *four viz*,

- i. Vulnerable
- ii. Rare
- iii. Intermediate
- iv. Threatened.

Endangered species which are on verge of extinction are called *threatened species*. Today most of the endangered species are found only in protected areas (PAs). Some examples of the species being tiger, rhino, elephant; bird species include siberian crane, great Indian bustard, florican, vultures; reptiles and amphibians. Habitat loss caused by human activity is causing threat to plants species like orchids. Over harvesting of medicinally important plants as ingredients in medical products or cosmetics is also threatening species. To protect endangered species India has created a wildlife protection act. Under this plants and animals are characterized according to the threat to their survival.



Biodiversity and its Threat

The species which are unique to a locality/region are called *endemic species*. Some species are found only in India and are thus endemic (restricted to our country). Some have very localized distribution and are considered highly endemic. Some species of this category being Indian wild ass, angular Kashmiri stag, golden langur, pigmy hog, etc.

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11.9 SUMMARY

1. Biodiversity is threatened by the sum of all human activities. It is useful to group threats into the categories of habitat destruction, invasion by non-native species, overhunting, pollution, and climate change.
2. Habitat loss presents the single greatest threat to world biodiversity, and the magnitude of this threat can be approximated from species-area curves and rates of habitat loss. The spread of non-native species threatens many local species with extinction, and pushes the world's biota towards a more homogeneous and widely distributed subset of survivors.
3. Climate change threatens to force species and ecosystems to migrate towards higher latitudes with no guarantee of suitable habitat or access routes.
4. Many species have gone extinct, some naturally and others because of human activities i.e. deforestation, desertification, and destruction of wetlands and coral reefs.
5. Habitats that are vulnerable to greater species extinction are referred to as fragile habitats. Coral reefs, oceanic islands, mountain tops, and habitat islands are considered as fragile habitats.

11.10 SELF-ASSESSMENT QUESTIONS

1. What are the demerits of domestication of selective animals? How can it destroy biodiversity?
2. Discuss with an example, how is poaching responsible for extinction of big animals
3. Discuss the harmful effects of invasive species with example.

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11.11 REFERENCES

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LESSON 12

BIODIVERSITY AND ITS CONSERVATION

NOTES

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Structure

- 12.1 Learning Objectives
- 12.2 *In-situ* and *Ex-situ* Conservation of Biodiversity
- 12.3 Conservation of cultivators and livestock breeds
- 12.4 Biological Diversity Act 2002
- 12.5 Need for conserving Biodiversity
- 12.6 Conservation of Biodiversity
- 12.7 National parks and sanctuaries
- 12.8 Natural Reserves
- 12.9 Wetlands
- 12.10 Hotspots of biodiversity
- 12.11 IUCN red data list
- 12.12 Summary
- 12.13 Self-Assessment Questions
- 12.14 References
- 12.15 Suggested Readings

12.1 LEARNING OBJECTIVES

After completing this lesson, you should be able to:

1. List various causes of biodiversity losses and discuss its major impact on the whole world.
2. Explain the need and concept of biodiversity conservation.
3. Enumerate the needs for *in-situ* conservation of species along with the various measures and mechanisms.
4. Appreciate the need for *ex-situ* conservation of species and the various mechanisms that are involved in this process of conservation.

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12.2 IN-SITU AND EX-SITU CONSERVATION OF BIODIVERSITY

In-situ conservation: It is the method of conserving all the living species, especially the wild and endangered species in their natural habitats and environment. In-situ conservation of biodiversity includes biosphere reserves, national parks, wildlife sanctuaries, etc. Habitat is protected with all the other species that is in nature. Biodiversity at all levels can be best preserved in-situ by setting aside wilderness as protected areas in national parks and wildlife sanctuaries with distinctive ecosystem included in the network.

India has 589 PAs (Protected Areas) of which 89 are national parks and 500 are wildlife sanctuaries. Over 100 PAs are created in Andaman and Nicobar to preserve the special island ecosystem. The great Himalayan national park is the largest sanctuary in the ecosystem and is the home of snow leopard. Dachigam sanctuary for hangul or Kashmiri stag; Kaziranga national park for animals like elephant, guar, wild boar and swamp deer, and birds like ducks, geese, pelicans and storks; Manas sanctuary for Golden langur, pigmy hog, and wild boar are some of the examples worth mentioning under in-situ conservation.

Ex-situ conservation: Conserving the species outside the natural habitat in a carefully controlled situation, such as botanical garden for plants or zoological parks for animals, expertise exists to multiply species under artificially managed condition. Germplasm is preserved in a gene bank for future needs, this is taken up for expensive endangered/extinct species. Care is taken to avoid inbreeding such that weak offspring would not develop. Breeding programmes in zoos provide animal needs including enclosures that simulate their wild habitat. Modern zoo's function is to breed endangered species as a conservation. Successful examples are:

1. Madras crocodile trust bank has successfully bred the 3 crocodiles. Here crocodiles lay two clutches of eggs in one year instead of one in wild.
2. Guchali zoo has bred pigmy hog.
3. Delhi zoo has bred the rare Manipur brow antlered deer.



The successful breeding programme also aims at reintroduction of the species into wild habitat with simultaneous removal of problems like poaching disturbances and man-made influences.

12.3 CONSERVATION OF CULTIVATORS AND LIVESTOCK BREEDS

Fifty years ago nearly 30,000 rice varieties were grown in India. Now only a few of these are cultivated. The new varieties being developed use the germplasm of these original types. But if all these traditional types vanish, it would be difficult to develop new disease resistant varieties for future. Use of varieties from gene banks have been expensive and risky. Farmers need to be encouraged to grow traditional varieties. Traditional breeds/varieties have to be encouraged for genetic variability. In contrast men interested in cash returns in short time wouldn't appreciate the benefits of growing indigenous varieties.

12.4 BIOLOGICAL DIVERSITY ACT 2002

Biological diversity is a national asset of a country, hence the conservation of biodiversity assumes greater significance. The first attempt to bring the biodiversity into the legal frame work was made by the Biodiversity bill 2000 which was passed by the Lok sabha on 2nd December 2002 and by the Rajya Sabha on December 2002.

Objectives:

1. To conserve the Biological Diversity.
2. Sustainable use of the components of biodiversity.
3. Fair and equitable sharing of benefits arising out of the use of the Biological Diversity.

A national biodiversity authority has been established by the Biodiversity Act 2002 to regulate act implementing rules 2004 has been operationalised since



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coming in to force. Regulating access as well as pushing the officially sponsored documentation of biological resources and traditional practices through people's diversity registers at the local and databases at the national levels, respectively. It further probes the extent to which the principles of conservation have been realised.

Provisions of Act:

1. Prohibition on transfer of Indian genetic material outside the country without specific approval of the Indian Government.
2. Prohibition of anyone claiming an (IPR) such as a patent over biodiversity or related knowledge without permission of Indian Government.
3. Regulation of collection and use of biodiversity by Indian national while exempting local communities from such restrictions.
4. Measures from sharing of benefits from use of biodiversity including transfer of technology, monetary returns, joint research and development, joint IPR ownership, etc.
5. Measures to conserve sustainable use of biological resources including habitat and species protection (EIP) of projects, integration of biodiversity into the plans, and policies of various departments and sectors.
6. Provisions for local communities to have a say in the use of their resources and knowledge and to charge fees for this.
7. Protection of indigenous or traditional laws such as registration of such knowledge.
8. Regulation of the use of the genetically modified organisms.
9. Setting up of National, state, and local Biodiversity funds to be used to support conservation and benefit sharing.
10. Setting up of Biodiversity Management Committees (BMC) at local village levels, State Biodiversity Boards at state level, and National Biodiversity Authority.



Functions of Authority:

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1. Advise the central Government on any matter concerning conservation of biodiversity. Sustainable use of its components and fair as well as equitable sharing of benefits arising out of the use of biological resources and knowledge.
2. Coordinate the activities of state biodiversity.
3. Provide the technical assistance and guidance to the state biodiversity boards.
4. Sponsor investigation and research.
5. Engage consultants for a specific period and not exceeding 3 years for providing technical assistance to the authority in the effective discharges of its functions.
6. Collect, compile, and publish technical and statistical data, manuals, codes or guides relating to conservation of biodiversity, sustainable use of its components and fair as well as equitable sharing of benefits arising out of the use of biological resource and knowledges.
7. Organise through mass media, a comprehensive programme regarding conservation of biodiversity, sustainable use of components, and fair and equitable sharing of benefits arising out of the use of biological resources and knowledge.
8. Plan and organize training of personal engagement or likely to be engaged programmes for the conservation of biodiversity and sustainable use of its components.
9. Prepare the annual budget of the authority including its own receipts as the devaluation from the central government provided that the allocation by the central government shall be operated in accordance with budget provisions approved by the central government.
10. Recommend creation of posts to the central government for effective discharge of the functions by the authority.
11. Approve the method of recruitment to the officers and servants of the authority.



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12. Take steps to build up data base and to create information and documentation system for biological resources and associated traditional knowledge through biodiversity. Register electronic databases to ensure effective management, promotion, and sustainable uses.
13. Give directions to the State Biodiversity Boards and the Biodiversity Management Committees in writing for effective implementation of the act.
14. Report to the central government about the functioning of the authority and implementation of the act.
15. Sanction grants to the State Biodiversity Board and Biodiversity Management Committees for specific purposes.
16. Take necessary measures including appointment of legal experts to oppose the grant of intellectual property right in any country outside India on any biological resource and associated knowledge obtained from India in an illegal manner.
17. Do other functions that may be assigned by the central government to direct from time to time.
18. Regulates the commercial utilisation or biosurvey and bio-utilisation of any biological resource by Indians.

Note: *International day for Biological Diversity – 22nd May*

12.5 NEED OF CONSERVING BIODIVERSITY

Why is biodiversity so important? Why should we care about it? What is the value of biodiversity? Maybe people don't understand the various roles biodiversity plays in our life, but certainly they know the importance of it. There are many factors that underlie the need to conserve biodiversity, such as:

1. Present and potential uses of the components of biological diversity especially as we have no way of knowing or predicting what will be of use in the future.
2. Biodiversity is essential to maintain the earth's life support systems that enable the biosphere to support human life.

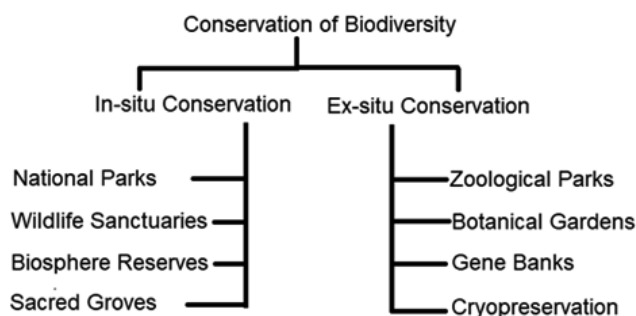


Fig. 1

It is ethically important to maintain all of the earth's biological diversity, including all the other extent (currently existing) life forms.

12.6 CONSERVATION OF BIODIVERSITY

Conservation needs different strategies, they can be species based or habitat based or ecosystem based. Some species are given importance at national level while some needs treatment at international level. Most of the conservation is done at *in-situ* and *ex-situ* conditions. In this lesson, we will discuss what these conditions mean, what is the difference between them, and what are the methods and techniques used. We have also described some important projects such as project tiger and how this project has helped in increasing tiger populations. Some techniques such as seed bank and tissue culture are also proving very helpful in conservation of plants which fulfill several of our needs.

In-situ conservation means “*on-site conservation*”. It is the process of protecting an endangered plant or animal species in its natural habitat, either by protecting or cleaning up the habitat itself or by defending the species from predators. The benefit of *in-situ* conservation is that it maintains the recovering populations in the surroundings where they have developed their distinctive properties.

Wildlife conservation is mostly based on *in-situ* conservation. This involves the protection of wildlife habitats. Also, sufficiently large reserves are maintained to enable the target species to exist in large numbers. The population size must be sufficient to enable the necessary genetic diversity to survive within the population.



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Ex-situ conservation means, literally “*off-site conservation*”. It is the process of protecting population of an endangered species of plant or animal by removing it from an unsafe or threatened habitat and placing it or a part of it, under the care of humans. While *ex-situ* conservation is comprised of some of the oldest and best known conservation methods known to human, it also involves newer, sometimes controversial laboratory methods. It is to be used as a last resort or as a supplement to *in-situ* conservation because it cannot recreate the habitat as a whole: the entire genetic variation of a species, its symbiotic counterparts, or those elements which over time might help a species to adapt to its changing surroundings. Furthermore, *ex-situ* conservation techniques are often costly. Plants and animals living in *ex-situ* breeding grounds have no natural defence to the diseases and pests that are new to the species.

12.7 NATIONAL PARKS AND SANCTUARIES

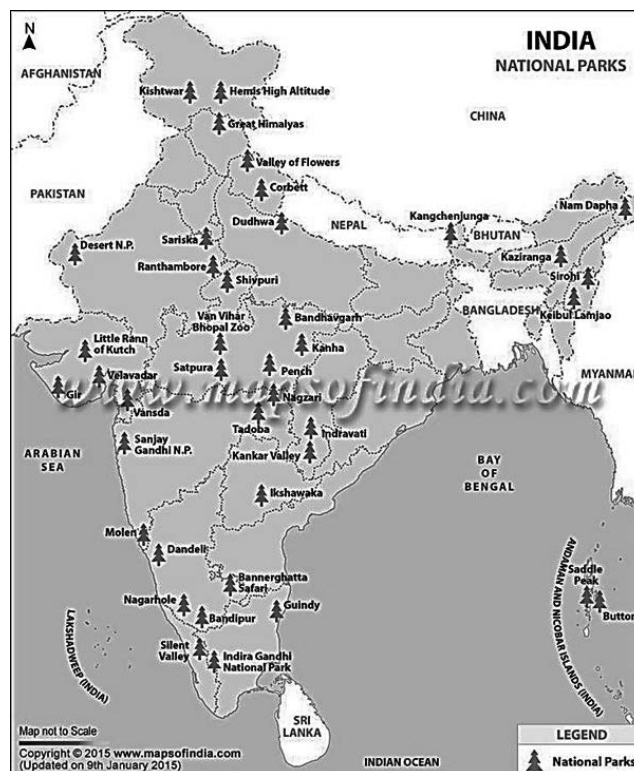


Fig. 2



Case study: Project Tiger

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Project tiger was launched on 1 April, 1973 as a centrally sponsored scheme of government of India to maintain viable population of the tigers and its natural habitat. The main objective of this scheme is to ensure that a viable population of tigers in India are preserved for scientific, economic, aesthetic, cultural, and ecological values and to also preserve the areas of biological importance as natural heritage for the benefit, education, and enjoyment of the people. Main objectives under the scheme include wildlife management and protection.

Initially, the project started with nine tiger reserves, covering an area of 16,339 sq.km with a population of 268 tigers. At present there are 27 tiger reserves covering an area of 37,761 sq.km. with a population of 2967 tigers. This amounts to almost 1.14% of the total geographical area of the country. The selection of reserves was guided by the representation of Eco typical wilderness areas across the biogeographic range of tiger distribution in the country. Project Tiger is undisputedly a custodian of major gene pool in the country. It is also a repository of some of the most valuable ecosystem and habitats for wildlife.

The Tiger Reserves are constituted on a '*core-buffer strategy*'. The core area is kept free of biotic disturbances and forestry operations, collection of minor forest produce, grazing, and human disturbances are not allowed within. However, the buffer zone is managed as a '*multiple use area*' with twin objectives of providing habitat supplement to spill over the population of wild animals from the core conservation unit and to provide site specific eco-developmental inputs to surrounding villages for relieving the impact on the core. No relocation is visualised in the buffer area and forestry operations. Non-Timber Forest Produce (NTFP) collection and other rights and concessions to the indigenous communities are permitted in a regulated manner to complement the initiatives in the core unit. The effective protection and intensive conservation measures inside the reserves have brought about considerable indescribable achievements. The project has been instrumental in mustering local support for conservation programme in general.



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Fig. 3

Botanical Gardens

Together, the world's 1500 botanic gardens (Fig. 8.4), arboreta, and national plant collections maintain the largest array of plant diversity outside of nature, and they have major, if often overlooked, potential as resource centres for conservation, education, and development. If the infrastructure and technical facilities of most of these institutions can be strengthened, they can conserve *ex situ* stocks of most of the world's endangered plant species. Already, individuals of an estimated 12,000 to 15,000 threatened species are being cultivated in botanic gardens and arboreta.

Zoological Parks

The basic philosophy behind the creation of zoological parks in modern times is to create an understanding of the environment and ecological balance of life, meaning strengthening the bond between people and the living earth. These zoological parks are no mere picnic spots. They are now centres for *ex-situ* wild life conservation and environmental education.

The history of modern zoos started around 200 years ago with the creation of the first public zoo. Since then every part of world has developed their own zoological parks with great diversity such as aquaria, bird-parks, private zoos and safari parks. The World Zoo Conservation Strategy concludes that the evolution



of zoo should continue to help the conservation of wildlife. There are several species of wildlife which would have been extinct today except for efforts by zoos and animal reserves.

Ex-Situ conservation of wild animals in zoo

Some important techniques used in *ex-situ* conservation are dealt here.

- 1. Captive breeding:** It is one of the important strategies used by both government and non-government organisations. Captive-breeding programmes of endangered and threatened species have become familiar programmes that strive to preserve biodiversity and species survival plans such as cheetah.
- 2. Embryo Storage and transfer technology:** Techniques for embryo transfer and artificial insemination, which have been developed for laboratory animals and farm animals, are potentially very useful for improving the reproductive potential of captive populations of endangered species. These kinds of techniques have been worked out mainly for mammals.
- 3. Artificial insemination:** It is another useful technology. Sperm can be frozen and used later, or transferred to another breeding facility to increase genetic diversity. Sometimes, the sperm can be added to the eggs in a dish and fertilisation will occur. In other cases (for example, horses) the sperm has to be injected into the egg. Few years ago, the black-footed ferret was down to six individuals, but artificial insemination has now been used to produce 16 kittens. Elephants and cheetahs have conceived, and a cheetah cub has been born following artificial insemination. Elephants have not bred naturally in captivity, so this method may be useful simply to make captive breeding possible.
- 4. Somatic cell cloning:** It holds some promise for propagating from one or a few survivors of an almost extinct species. This was first done with domestic sheep at the Roslyn Institute in Edinburgh (from University of Virginia) but since then it has been done with other mammals. It has already been used to rescue a rare breed of cattle that had been reduced to a single old female ("Lady") and some frozen sperm. Granulosa cells (somatic cells in the ovary) from Lady were fused with enucleated eggs (lacking

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- DNA) from a different breed, and the resulting eggs were implanted into an Angus cow (a common breed). The first calf born from these cells is genetically identical to the Lady, as expected, although her markings were slightly different.
5. **Fostering:** Many egg-laying animals (i.e. birds and reptiles) are capable of producing many more eggs than they can rear. This raises the possibility of collecting the extra eggs and hatching and rearing the animals in captivity with a foster parent, then using them to supplement wild populations. It has worked extremely well with some birds, particularly the peregrine falcon, which is now doing so well that the fostering programme is being phased out. Rearing of whooping cranes has also been successful, and the species recovered from a population of 21 birds in 1941 to over 300 in 1996
 6. **Translocations:** Sometimes conservation of faunal species involves or necessitates translocation of animals. This means the movement of individuals from its natural habitat or captivity to another habitat. Translocations are carried out in connection with introductions or reintroductions, and should be handled with extreme caution.
 7. **Introduction:** This involves the translocation of a species (from its natural habitat or from captivity) into an area outside its historical distribution. Such species would then become an “*exotic*” to the area. This should be handled with extreme care and needs. Extensive study of the habitat and the behaviour and social organization of the species to be introduced has to be done to ensure that the species has a good chance of adapting to the habitat.
 8. **Reintroduction:** This involves the translocation of a species (from its natural habitat or from captivity) into an area within its historical distribution, either to boost existing populations or to establish new populations when the original population has died out. This too should be handled with extreme care and needs extensive study of the habitat and the behaviour and social organization of the species to be reintroduced.
 9. **Seed Bank:** The preservation of plant germplasm in seedbanks, (or gene banks), is one of the techniques of *ex-situ* conservation of plant species.



Seeds have a natural dormancy feature, which allows for their suspended preservation for long periods of time with little damage, provided the conditions are favourable. Banking dormant seeds enables to keep genetically representative samples of rare and endangered plant species as a kind of “genetic insurance”.

- 10. Seeds Storing:** Storing germplasm in seedbanks is both inexpensive and space efficient. It allows preservation of large populations with little genetic erosion. Seedbanks also offer good sources of plant material for biological research, and avoid disturbance or damage of natural populations.

12.8 NATURAL RESERVES

The nature reserves are the important area for the conservation of biodiversity. The growing destruction of biodiversity re-emphasises the valuable contribution of natural reserves. These areas are resourceful and useful means to deal with biodiversity losses and help in buffering society from climatic effects and maintains the critical ecosystem services of the society.

Biosphere Reserves are internationally recognised areas established to promote and demonstrate a balanced relationship between humans and the biosphere. They highlight the value of nature conservation within a particular natural region and reconcile the conservation of biological diversity with sustainable use. Consequently they are ideally suited to meet today’s conservation needs when human populations are increasing and the practicality of leaving aside large areas to protect pristine natural wild lands is decreasing, despite the fact that more people than ever before are dependent on wild species and natural ecosystems for their well-being.

The programme of Biosphere Reserve was initiated under the *Man and Biosphere (MAB)* programme by UNESCO in 1971. Biosphere Reserves programme is intended to conserve representative ecosystems as opposed to only species or habitat conservation. It provides in-situ conservation under natural conditions, long-term conservation of plants, animals, and microorganisms.



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12.9 WETLANDS

India's wetlands are distributed in different geographical regions ranging from the cold arid zone of Ladakh to the wet humid climate of Imphal; the warm arid zone of Rajasthan to the tropical monsoonal Central India, and the wet humid zone of the Southern peninsula.



Fig. 4

All over the world, a wetland is considered to be an important part as it:

1. is particularly a good representative example of a natural or near natural wetland, characteristic of the appropriate biogeographic region.
2. plays a substantial hydrological, biological, or ecological role in the natural functioning of a major river basin or coastal system.
3. is a specific type of wetland, rare or unusual.
4. supports an appreciable assemblage of rare, vulnerable or endangered species, or subspecies of plants or animals.

Importantly, wetlands are habitats for a wide variety of plant and animal life. Most important of these is the waterfowl as wetlands are shallow water.



Apart from harbouring birds, wetlands are also a nursery ground for several species of fish and shell fish and a wide variety of aquatic organisms. For example, Chilka in Odisha has dolphins that move around in the area where the lake meets the sea. Coastal wetlands especially being an ecotone between the sea and freshwater, and/or freshwater and terrestrial habitats have high species diversity. Ecologically, too, wetlands perform important functions. They regulate the water regime, acts as natural filters, and display a marvelous nutrient dynamics. As an ecosystem, wetlands are useful for nutrient recovery and cycling, releasing excess nitrogen, deactivating phosphates, removing toxins, chemicals and heavy metals through absorption by plants and also in the treatment of waste water.

Furthermore, coastal wetlands with their unique mangroves are a natural bulwark against erosion by the sea. The possible threat of rise in sea level is universally dreaded. One immediate prevention for this possible threat, as has been suggested by experts, would be the plan of a network of mangroves. In fact, mangrove wetlands of India and Bangladesh act as buffers against the devastating storms of the Bay of Bengal. Wetlands thus help in mitigating floods, recharging aquifers, and reducing surface run-off as well as the consequent erosion.



Fig. 5: Mangroves acting as a bulwark for the environment

In the context of the environment too, wetlands play a very important role. They protect and improve the quality of water and keep the local weather moderate. Using wetlands for water quality improvement has been tried in cold climates. Wetlands in urban periphery are natural receptacles for waste water and can harness effectively the nutrients available in the waste through fisheries and agriculture.



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Natural Wetlands of India

Most of the natural wetlands of India are connected with the river systems of the North and South. On the other hand, the various multi-purpose projects launched to harness river systems have provided a number of wetlands. E.g., Harike Barrage at the confluence of the Beas and Sutlej in Punjab, Bhakra Nangal Dam in Punjab, and the Kosi Barrage on the Bihar-Nepal border. Besides these, we also have a network of lakes – natural as well as manmade, for example, Kabar lake, Chilka lake, Pichola Complex and Sukhna lake, etc. In addition of these, there are 6,740 square kilometres of mangroves. The major concentrations of mangroves in the country are the Sunderbans and the Andaman and Nicobar Islands, which holds 80% of the mangroves in India. Rest of them are in Odisha, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra, Gujrat, and Goa. The two wetlands from India which found place in first International Convention on Wetlands held at Ramsar in Iran in February 1971 were Chilka and Bharatpur. Currently there are 26 Ramsar sites in India, covering most of the important wetlands.

12.10 HOTSPOTS OF BIODIVERSITY

It has been observed that enormous number of species that exists on Earth is concentrated in specific regions. Two-third of the species are found in geographical regions within tropical areas (area between Tropic of Cancer and Tropic of Capricorn). This indicates that the species are not evenly distributed on this planet. Tropical region is known to possess high species richness (number of species living in a specific location). Biodiversity conservation requires identification of such regions that are habitat of vast number of species. Geographical regions that support number of species and threatened by destruction are known as hotspots.

Hotspots of biodiversity are those places on earth which have high biodiversity and are threatened by human habitation. The term ‘Hotspot of Biodiversity’ was given by the British biologist Norman Myers in 1988. Meyer defined hotspot as “**a biogeographic region characterised by exceptional levels of endemism and by serious level of habitat loss**”.



Hotspots of biodiversity are identified based on following two criterias:

1. Area must support **at least 1,500 vascular plants (> 0.5% of the world's total) as endemics**. This indicates that the area must have a high percentage of plant life found nowhere else on the planet. A hotspot is considered as **irreplaceable**.
2. Area must include **30% or less of its original natural vegetation** which indicates that it must be threatened. At present, 36 hotspots have been identified in the world. Though these 36 hotspots together represent only 2.4% of the Earth's land surface but they are crucial in supporting more than half of the world's plant species as endemics and nearly 43% of bird, mammal, reptile, and amphibian species as endemics. Further, these 36 hotspots provide ecosystem services and livelihood to more than 2 billion people on Earth. Out of these 36 hotspots, 4 hotspots are found in India. These four hotspots are:
 - **Himalaya:** Includes the entire Indian Himalayan region (and that falling in Pakistan, Tibet, Nepal, Bhutan, China, and Myanmar).
 - **Indo-Burma:** Includes entire north-eastern India, except Assam and Andaman group of Islands (as well as Myanmar, Thailand, Vietnam, Laos, Cambodia, and southern China).
 - **Sundalands:** Includes Nicobar group of Islands (as well as Indonesia, Malaysia, Singapore, Brunei, and Philippines).
 - **Western Ghats and Sri Lanka:** Includes entire Western Ghats (and Sri Lanka).

12.11 IUCN RED LIST

IUCN (International Union for Conservation of Nature and Natural Resources) is an international organization which is responsible for assessment of species all across the world. IUCN conducts field survey to monitor species in every country. Based on the survey, IUCN classify plants and animals in different categories. This information about current status of species is published by IUCN in '**Red Data List**' publication.



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12.12 SUMMARY

1. Major impacts of biodiversity loss are steady increase in atmospheric CO₂ level, adverse effects on local climate and water flow, reduction of genetic diversity, extinction of species, and loss of livelihoods.
2. In-situ conservation meaning on-site conservation and ex-situ conservation meaning off-site conservation are the two important ways of conservation of species. Wildlife conservation is mostly based on in-situ conservation. Ex-situ conservation is a man's efforts to sustain and protect the environment and ex-situ conservation is used when species extinction is imminent.
3. In-situ conservation of species is generally operated in places like National Parks and Wildlife Sanctuaries, Wetlands, Biosphere Reserves, Tiger Reserves, and Elephant Reserves.
4. Ex-situ conservation sites are Botanical Gardens and Zoological Parks.
5. Seed banks and tissue culture are important methods for plant conservation.
6. Nature reserves are hubs of biodiversity and they should be left as such so that ecosystem can function properly.

12.13 SELF-ASSESSMENT QUESTIONS

1. Why is biodiversity important for human beings? Make a list of advantages from biodiversity.
2. Describe how is global climate change harming biodiversity on the earth.
3. Why is biodiversity important? Discuss some of its values and indicate the ones you favour the most.
4. Describe each of the five major threats to biodiversity. Give an example of a species affected by each of these threats.
5. Differentiate between *in-situ* and *ex-situ* conservation. What is the ultimate goal of captive breeding? Why is it best used only as a last resort?



Biodiversity and Its Conservation

6. Visit a local zoo or botanical garden. What are the activities conducted there to promote biological conservation? List them.
7. What are the advantages of tissue culture ?
8. Write a note on Biosphere Reserves.

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LESSON 13

SUSTAINABLE DEVELOPMENT & SUSTAINABILITY

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Structure

- 13.1 Learning Objectives
- 13.2 Introduction
- 13.3 Types of Sustainability
 - 13.3.1 Environmental Sustainability
 - 13.3.2 Economic Sustainability
 - 13.3.3 Social Sustainability
- 13.4 Summary
- 13.5 Self-Assessment Questions
- 13.6 References

13.1 LEARNING OBJECTIVES

- Environmental communication and public awareness (e.g., Sustainable Development and human communities).
- Role of public awareness for Sustainable development.

13.2 INTRODUCTION

Sustainable Development is defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It also looks at the equity between countries and continents, races and classes, gender and ages. It includes social development and economic opportunity on one hand and the requirements of environment on the other. It is based on improving the quality of life for all, especially the poor and deprived within the carrying capacity of the supporting ecosystems. It is a process which

Self-Instructional
Material

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leads to a better quality of life while reducing the impact on the environment. Its strength is that it acknowledges the interdependence of human needs and environmental requirements.

To ensure sustainable development, “any activity that is expected to bring about economic growth must also consider its environmental impacts so that it is more consistent with long term growth and development.” Many ‘development projects’, such as dams, mines, roads, industries, and tourism development have severe environmental consequences that must be studied before they are even begun. Thus for every project, in a strategy that looks at sustainable development, there must be a scientifically and honestly done EIA, without which the project must not be cleared.

Large dams, major highways, mining, industry, etc. can seriously damage ecosystems that support the ecological health of a region. Forests are essential for maintaining renewable re- sources, reducing carbon dioxide levels, and maintaining oxygen levels in the earth’s atmosphere. Their loss impairs future human development. Loss of forests depletes biodiversity which has to be preserved to maintain life on earth. Major heavy industries if not planned carefully lead to environmental degradation due to air and water pollution and generate enormous quantities of waste that lead to long term environmental hazards. Toxic and Nuclear wastes can become serious economic problems as getting rid of them is extremely costly. Thus the economic benefits of a project must be weighed against the possible environmental costs before a project is permitted.

We as citizens of our nation, and increasingly as citizens of one common future at the global level, must constantly monitor the pattern of development. If we see that a development project or an industry is leading to serious environmental problems, it is our duty to bring this to the attention of authorities such as the local administration, the Forest Department, or the Pollution Control Board to look into the issue. Further, if new development projects are being planned in and around the place where we live it is our duty to see that this is brought about in accordance with environmental safeguards. While we all need to think globally, we need to act locally. We have to see to it that we change development from its present mandate of rapid economic growth without a thought for future ecological integrity, to a more sustainable ecologically appropriate strategy.



Fig. 1

13.3 TYPES OF SUSTAINABILITY

13.3.1 Environmental Sustainability

Environmental sustainability focuses on the conservation of biodiversity without foregoing economic and social progress. The foundations of environmental sustainability are safeguarding water, saving energy, reducing waste, using recyclable packaging, limiting or eliminating the use of plastics, using sustainable transport, reusing paper, and protecting flora and fauna.

A great example of environmental sustainability is the Swedish city of Stockholm, which is noted for investing in sustainable infrastructure, its low emissions, and for having excellent air quality with pollution levels below average. The city has thus achieved a balance between economic development and environmental protection.



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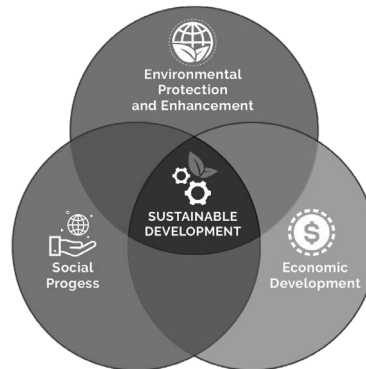


Fig. 2

Another example is the Bimbo Group, which aims to reduce its environmental impact through regenerative agriculture with zero carbon emissions, the use of renewable energies, and the reduction of waste.

13.3.2 Economic Sustainability

When a company is set up, a structure is created that involves expenses and revenues. Once a balance is struck between both factors, the company earns profit. Economic sustainability refers to the organisation's ability to manage its resources and responsibly generate profits in the long term.

An example of this type of sustainability is the company Unilever, which in the year 2010 rolled out a strategy to achieve a balance between sustainability and the company's economic performance. To do so, it implemented several measures, such as increasing package recycling, promoting the use of recycled materials, and responsible consumption awareness campaigns.

Likewise, we can also look to the case of the Suez company, which in its Sustainable Development Report 2020 reveals that it has reduced its emissions related to electricity consumption by 95%, by purchasing and generating renewable energy; that it has implemented energy efficiency measures and that, additionally, in terms of the conservation of natural habitats, some 81.5% of its facilities are free from pesticides and crop protection chemicals.



13.3.3 Social Sustainability

In any community in which economic activities are carried out in a specific environment, we find three interconnected forms of sustainability: environmental, economic, and social. However, social sustainability in particular has the goal of strengthening the cohesion and stability of specific social groups.

The company CEMEX, for example, is working to contribute to the social development of communities. Thus, the company offers decent housing through self-building programmes and loans with favourable access conditions to those who are most in need.

The Gigante Group is another example, since, via the Gigante Foundation, it contributes funds and resources to a range of social causes, such as school materials for collaborators, and grants to improve visual health.

13.4 SUMMARY

Creativity, technology, and financial resources from all of the society is necessary to achieve the sustainable development and it is clear from the lesson that sustainable development is a broad term to describe policies, projects, and investments that provide benefits today without sacrificing environmental, social, and personal health in the future. So sustainable development is an organising principle that aims to meet human development goals while also enabling natural systems to provide necessary natural resources and ecosystem services to humans. The desired result is a society where the living conditions and resources meet human needs without undermining the planetary integrity and stability of the natural system. Sustainable development tries to find a balance between economic development, environmental protection, and social well-being. The Brundtland Report in 1987 defines sustainable development as “development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs”. The concept of sustainable development nowadays has a focus on economic development, social development, and environmental protection for future generations.

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13.5 SELF-ASSESSMENT QUESTIONS

1. Discuss the role of environmental communication and public awareness in environment protection.
2. In your view, how are the environmental and ecological rights related to democracy and development of India ? Explain.
3. With appropriate examples, discuss the contribution of public in protecting the environment and raising environmental awareness in India.

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LESSON 14

GLOBAL ENVIRONMENTAL ISSUES

NOTES

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Structure

- 14.1 Introduction
- 14.2 Climate change
- 14.3 Global warming
- 14.4 Acid rain
- 14.5 Ozone layer depletion:
- 14.6 Nuclear Accidents and Nuclear Holocaust
- 14.7 Summary
- 14.8 Self-Assessment Questions
- 14.9 References

14.1 INTRODUCTION

Human population size has grown enormously over the last hundred years. This means there will be increase in demand for food, water, home, electricity, roads, automobiles, and numerous other commodities. These increased demands are exerting tremendous pressure on our natural resources, and are also contributing to the pollution of air, water, and soil. The need of the hour is to check the degradation and depletion of our precious natural resources and pollution without halting the process of development.

14.2 CLIMATE CHANGE

The average temperature in many regions has been increasing in recent decades. The global average surface temperature has increased by $0.6^{\circ} + 0.2^{\circ}$ C over the last century. Globally, 1998 was the warmest year and the 1990s the warmest decade

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on record. Many countries have experienced increase in rainfall, particularly in the countries situated in the mid to high latitudes.

In some regions, such as parts of Asia and Africa, the frequency and intensity of droughts have been observed to increase in recent decades. Episodes of El Niño, which creates great storms, have been more frequent, persistent, and intense since mid 1970s compared to the previous 100 years. All these are signs that the earth is sick. Its climate is changing, making it more difficult for mankind to survive. The earth is continuously losing its ability to balance itself due to the imbalances created by human activities.

Projections of future climate change are derived from a series of experiments made by computer based global climate models. These are worked out on estimates of aspects such as future population growth and energy use. Climatologists of the Intergovernmental Panel on Climate Change (IPCC) have reviewed the results of several experiments in order to estimate changes in climate in the course of this century. These studies have shown that in the near future, the global mean surface temperature will rise by 1.4° to 5.8°C. Warming will be greatest over land areas, and at high latitudes. The projected rate of warming is greater than has occurred in the last 10,000 years. The frequency of weather extremes is likely to increase leading to floods or drought. There will be fewer cold spells but more heat waves. The frequency and intensity of El Niño is likely to increase. Global mean sea level is projected to rise by 9 to 88 cm by the year 2100. More than half of the world's population now lives within 200km of the coastline. They are likely to be seriously impacted by an ingress of salt water and by the rising sea. Some of the most vulnerable regions are the Nile delta in Egypt, the Ganges-Brahmaputra delta in Bangladesh, and many small islands including the Marshall Islands and the Maldives (WHO, 2001).

Human societies will be seriously affected by extremes of climate such as droughts and floods. A changing climate would bring about changes in the frequency and/or intensity of these extremes. This is a major concern for human health. To a large extent, public health depends on safe drinking water, sufficient food, secure shelter, and good social conditions. All these factors are affected by climate change. Fresh water supplies may be seriously affected, reducing the availability of clean water for drinking and washing during drought as well as



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floods. Water can be contaminated and sewage systems may be damaged. The risk of spread of infectious diseases such as diarrhoeal diseases will increase. Food production will be seriously reduced in vulnerable regions directly and also indirectly through an increase in pests and plant or animal diseases. The local reduction in food production would lead to starvation and malnutrition with long-term health consequences, especially for children. Food and water shortages may lead to conflicts in vulnerable regions with serious implications for public health. Climate change related impacts on human health could lead to displacement of a large number of people, creating environmental refugees and leading to further health issues.

Changes in climate may affect the distribution of vector species (e.g. mosquitoes) which in turn will increase the spread of disease, such as malaria and filariasis, to new areas which lack a strong public health infrastructure. The seasonal transmission and distribution of many diseases that are transmitted by mosquitoes (dengue, yellow fever) and by ticks (Lyme disease, tick-borne encephalitis) may spread due to climate change.

14.3 GLOBAL WARMING

The term *Environment* is derived from the French word '*Environ*' which literally means 'surrounding'. Anything and everything which surrounds us i.e. all living beings or biotic components (microbes, plants, and animals) and non-living or abiotic components (air, water, sunlight, etc.) present in nature forms the environment. The Environmental Protection Act of 1986 defines the Environment as "environment includes water, air, and land and the inter-relationship which exists among and between water, air and land, and human beings, other living creatures, plants, micro-organism, and property". Interactions between the biotic and abiotic components lead to a functional ecosystem and sustainable life on the planet earth. We get all the basic goods and services (clean air and water, food, fodder, medicines, raw materials for industries, tourism, etc.) from the environment. It is a well-known fact that anthropogenic activities and unsustainable consumption of natural resources by the human race have significantly damaged the environment and mother earth and the degradation



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is still going on at a fast pace. Therefore, it is our responsibility to protect the environment from getting degraded and polluted. Environmental education is indispensable for creating environmental awareness which ultimately will lead to environmental conservation.



Fig. 1

About 75% of the solar energy reaching the Earth is absorbed on the earth's surface which increases its temperature. The rest of the heat radiates back to the atmosphere. Some of the heat is trapped by greenhouse gases, mostly carbon dioxide. As carbon dioxide is released by various human activities, it is rapidly increasing. This is causing global warming.

The average surface temperature is about 15°C . This is about 33°C higher than it would be in the absence of the greenhouse effect. Without such gases most of the Earth's surface would be frozen with a mean air temperature of -18°C .

Human activities during the last few decades of industrialisation and population growth have polluted the atmosphere to the extent that it has begun to seriously affect the climate. Carbon dioxide in the atmosphere has increased by 31% since pre-industrial times, causing more heat to be trapped in the lower atmosphere. There is evidence to show that carbon dioxide levels are still



increasing. Many countries have signed a convention to reduce greenhouse gases under the United Nations Convention on climate change. Current international agreements are however not still effective to prevent the significant changes in climate and a rise in sea levels.

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14.4 ACID RAIN

When fossil fuels such as coal, oil, and natural gas are burned, chemicals like sulfur dioxide and nitrogen oxides are produced. These chemicals react with water and other chemicals in the air to form sulfuric acid, nitric acid, and other harmful pollutants like sulphates and nitrates. These acid pollutants spread upwards into the atmosphere and are carried by air currents to finally return to the ground in the form of acid rain, fog, or snow. The corrosive nature of acid rain causes many forms of environmental damage. Acid pollutants also occur as dry particles and gases, which when washed from the ground by rain, add to the acids in the rain to form a more corrosive solution. This is called *acid deposition*.

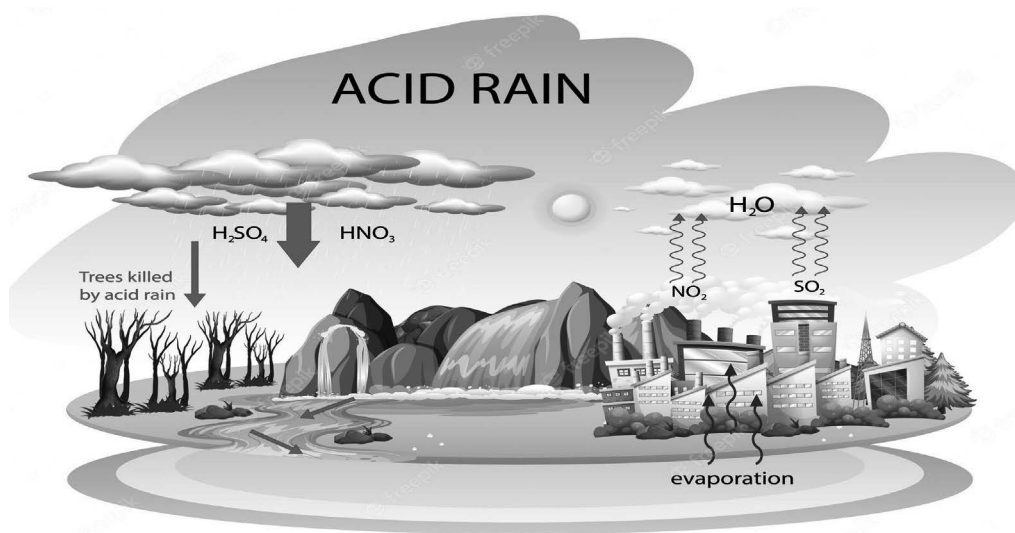


Fig. 2

Damage from acid rain is widespread in North America, Europe, Japan, China, and Southeast Asia. In the US, coal burning power plants contribute to

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about 70% of sulfur dioxide. In Canada, oil refining, metal smelting, and other industrial activities account for 61% of sulfur dioxide pollution. Motor vehicle exhaust fumes are the main source of nitrogen oxides. The acids in acid rain chemically react with any object they come in contact with. Acids react with other chemicals by giving up hydrogen atoms.

Effects:

Acid rain is known to cause following widespread environmental damage:

1. Acid rain dissolves and washes away nutrients in the soil which are needed by plants. It can also dissolve naturally occurring toxic substances like aluminium and mercury, freeing them to pollute water or poison plants.
2. Acid rain indirectly affects plants by removing nutrients from the soil in which they grow. It affects trees more directly by creating holes in the waxy coating of leaves, causing brown dead spots which affect the plant's photosynthesis. Such trees are also more vulnerable to insect infestations, drought, and cold. Spruce and fir forests at higher elevations seems to be most at risk. Farm crops are less affected by acid rain than forests.
3. Acid rain that falls or flows as ground water to reach rivers, lakes, and wetlands causes the water in them to become acidic. This affects plant and animal life in aquatic ecosystems.
4. Acid rain also has far reaching effects on wildlife. By adversely affecting one species, the entire food chain is disrupted, ultimately endangering the entire ecosystem. Different aquatic species can tolerate different levels of acidity. For instance clams and mayflies have a high mortality when water has a pH of 6.0, while frogs can tolerate more acidic water, although with the decline in supply of mayflies, frog populations may also decline. Land animals that are dependent on aquatic organisms are also affected.
5. Acid rain and dry acid deposition damages buildings, automobiles, and other structures made of stone or metal. The acid corrodes the materials causing extensive damage and ruins historic buildings. For instance the Parthenon in Greece and the Taj Mahal in India have been affected by acid rain.



6. Although surface water polluted by acid rain does not directly harm people, the toxic substances leached from soil can pollute water supply. Fish caught in these waters may be harmful for human consumption. Acid along with other chemicals in the air produces urban smog, which causes respiratory problems.

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Solutions:

The best way to stop the formation of acid rain is to reduce the emissions of sulfur dioxide and nitrogen oxides into the atmosphere. This can be achieved by using less energy from fossil fuels in power plants, vehicles, and industry. Switching to cleaner burning fuels is also a way out. For instance, using natural gas which is cleaner than coal, using coal with lower sulfur content, and developing more efficient vehicles. If the pollutants have already been formed by burning fossil fuels, they can be prevented from entering the atmosphere by using scrubbers in smokestacks in industry. These spray a mixture of water and limestone into the polluting gases, recapturing the sulfur. In catalytic converters, the gases are passed over metal coated beads that convert harmful chemicals into less harmful ones. These are used in cars to reduce the effects of exhaust fumes in the atmosphere. Once acid rain has affected soil, powdered limestone can be added to the soil by a process known as liming to neutralize the acidity of the soil.

14.5 OZONE LAYER DEPLETION

Ozone is formed by the action of sunlight on oxygen. It forms a layer 20 to 50kms above the surface of earth. This action takes place naturally in the atmosphere, but is very slow. Ozone is a highly poisonous gas with a strong odour. It is a form of oxygen that has three atoms in each molecule. It is considered as a pollutant at ground level and constitutes health hazard by causing respiratory ailments like asthma and bronchitis.



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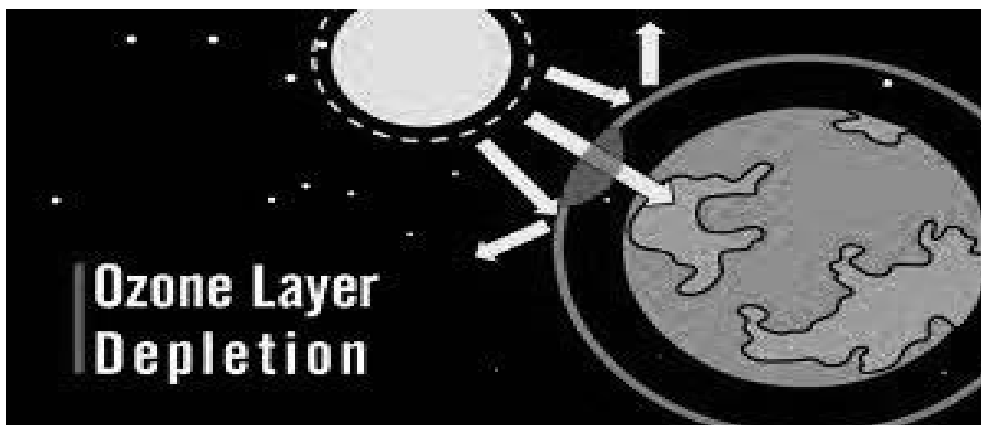


Fig. 3

It also causes harm to vegetation and leads to a deterioration of certain materials like plastic and rubber. Ozone in the upper atmosphere however, is vital to all life as it protects the earth from the sun's harmful ultraviolet radiations. The ozone layer in the upper atmosphere absorb the sun's ultraviolet radiations, preventing it from reaching the earth's surface.

This layer in the atmosphere protects life on earth from the dangerous UV radiations from the sun. In the 1970s, scientists discovered that chemicals called chlorofluorocarbons or CFCs, which were used as refrigerants and aerosol spray propellants, posed a threat to the ozone layer. The CFC molecules are virtually indestructible until they reach the stratosphere, where UV radiation breaks them down to release chlorine atoms. The chlorine atoms react with ozone molecules which break down into oxygen molecules, which do not absorb UV radiations. Since the early 1980s, scientists detected a thinning of the ozone layer in the atmosphere above Antarctica. This phenomenon is now being detected in other places as well including Australia. Although the use of CFCs has been reduced and now banned in most countries but other chemicals and industrial compounds such as bromine, halocarbons, and nitrous oxides from fertilizers may also attack the ozone layer.

The destruction of the ozone layer is seen to cause increased cases of skin cancer and cataracts. It also causes damage to certain crops and to plankton, thus affecting nature's food chains and food webs. This in turn causes an increase in carbon dioxide due to the decrease in vegetation.



With the signing of the Montreal Protocol in 1987, a treaty for the protection of the ozone layer, the use of CFCs was to be banned by the year 2000. After 2000, the ozone layer is expected to recover slowly over a period of about 50 years.

14.6 NUCLEAR ACCIDENTS AND NUCLEAR HOLOCAUST

Nuclear energy was researched and discovered by man as a source of alternate energy which would be clean and cheap compared to fossil fuels, and although this did happen, along with the benefits of nuclear energy came its down-falls. In the short history of nuclear energy there have been accidents that have surpassed any natural calamity or other energy source extraction in their impacts. A single nuclear accident can cause loss of life, long-term illness, and destruction of property on a large scale for a long period of time. Radioactivity and radioactive fall-out leads to cancer, genetic disorders, and death in the affected area for decades after, thus affecting all forms of life for the generations to come

Nuclear holocaust:

The use of nuclear energy in war has had devastating effects on man and earth. The Hiroshima and Nagasaki incident during World War II, the only use of nuclear power in war is one of the worst disasters in the history. In 1945, the United States dropped atomic bombs in Japan over the towns of Hiroshima and Nagasaki. These two atomic bombs killed thousands of people, left many thousands injured and devastated everything for miles around. The effects of the radiation from these nuclear bombs can still be seen today in the form of cancer and genetic mutations in the affected children and survivors of the incident.

14.7 SUMMARY

1. Environmental problem emerges out of the complex human environment interaction. Most of the environmental problems are trans-boundary in nature.



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2. Major causative factors responsible for today's environmental problems are industrialisation, urbanisation, explosive population growth, intensification of agriculture, rising use of fossil fuel based energy, and transportation to name a few.
3. Global warming refers to the rise of atmospheric temperature and consequent changes in the radiation balance mainly due to human action.
4. Ozone layer depletion is happening due to certain recent human activities which have injected certain chemicals in the stratosphere which consume ozone and reduce its concentration. Depletion is mainly caused by chlorofluorocarbons, halons, methyl chloroform, and carbon tetra chlorides. Climate change in today's context is mainly due to human activities. There are certain parameters that provide indication of warming of climate. Some of these indicators are (i) increasing temperatures over land and ocean surfaces; (ii) melting glacial ice and sea ice; (iii) rising sea level; and (iv) Increasing humidity.

14.8 SELF-ASSESSMENT QUESTIONS

1. Identify and list major global environmental issues.
2. Define and correlate global warming with green house effect.
3. Enumerate the major effects of global warming on living and non-living.
4. What are the components of the environment?
5. Briefly explain the causes of acid rain.
6. Comment on the major causes of desertification.
7. Explain the cause and effects of ozone-layer depletion.



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LESSON 15

ENVIRONMENTAL LAW & ENVIRONMENTAL LEGISLATIONS IN INDIA

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Structure

- 15.1 Learning Objectives
- 15.2 Introduction
- 15.3 The Indian Wildlife (Protection) Act, 1972
- 15.4 The Water (Prevention and Control of Pollution) Act, 1974
- 15.5 Forest (Conservation) Act, 1980
- 15.6 The Air (Prevention and Control of Pollution) Act, 1981
- 15.7 The Environment (Protection) Act 1986
- 15.8 Summary
- 15.9 Self-Assessment Questions
- 15.10 References

15.1 LEARNING OBJECTIVES

1. Gain knowledge about important environmental legislations enacted in India.
2. To learn the traditional wisdom and indigenous practices related to environmental protection.
3. To identify the constitutional provisions about the environment in the Indian constitution.

15.2 INTRODUCTION

In India, environmental protection and improvement has been in practice since ancient times. In ancient India, simple but quite effective laws were in place to protect and conserve environment. The long history of environmentalism in India

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is signified with the passage and codification of several acts, e.g. The Indian Motor Vehicle Act, The Factories Act, The Indian Forest Act, The Mines and Minerals (Regulation and Development) Act, The Industries (Development and Regulation) Act, The Forest (Conservation) Act, The Merchant Shipping Act, etc. The present day constitution of India allows the state to protect and improve the environment in order to safeguard public health, forests, and wildlife.

15.3 THE INDIAN WILDLIFE (PROTECTION) ACT, 1972

The Wildlife Act provides the State Wildlife Advisory Boards provisions to regulate hunting and trade of wild animals, birds, and their products. The State Wildlife Advisory Boards can also establish sanctuaries and national parks and judicially impose penalties for violating the Act. This Act was amended in 1982 to introduce provisions for the capture and transportation of wild animals to scientific institutions and bodies for management of animal populations. In 1991, the parent Act was comprehensively amended to insert special chapters dealing with the protection of specified plants and the regulation of zoos.

The major activities and provisions in the Act can be summed up as follows:



Fig.1



1. It defines the wildlife related terminology.
2. Appointment of Wildlife Advisory Board, wildlife warden, definition of their powers and duties, etc.
3. Listing of endangered wildlife species and provisions to prohibit their hunting.
4. Protection to some endangered plants like Bed dome cycad, Blue Vanda, Ladies slipper orchid, pitcher plant, etc.
5. Constitution of Central Zoo Authority.
6. Trade and commerce in some wildlife species with license for sale, possession, transfer, etc.
7. Ban on the trade or commerce in scheduled animals.

It provides for legal powers to officers and punishment to offenders. Several conservation projects for individual endangered species like lion (1972), tiger (1973), crocodile (1974), and brown antlered deer (1981) were started under this act. This Act is adopted by all Indian states.

The term Environment is derived from the French word '*Environ*' which literally means 'surrounding'. Anything and everything which surrounds us i.e. all living beings or biotic components (microbes, plants, and animals) and non-living or abiotic components (air, water, sunlight, etc.) present in nature forms the environment. The Environmental Protection Act of 1986 defines the Environment as "environment includes water, air, and land and the inter-relationship which exists among and between water, air, and land, and human beings, other living creatures, plants, micro-organism, and property". Interactions between the biotic and abiotic components lead to a functional ecosystem and sustainable life on the planet earth. We get all the basic goods and services (clean air and water, food, fodder, medicines, raw materials for industries, tourism, etc.) from the environment. It is a well-known fact that anthropogenic activities and unsustainable consumption of natural resources by the human race have significantly damaged the environment and mother earth and the degradation is still going on at a fast pace. Therefore, it is our responsibility to protect the environment from getting degraded and polluted. Environmental education is indispensable for creating environmental awareness which ultimately will lead to environmental conservation.



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15.4 THE WATER (PREVENTION AND CONTROL OF POLLUTION) ACT, 1974

The Water (Prevention and Control of Pollution) Act, 1974, defines water pollution as ‘the direct or indirect discharge of sewage, industrial effluents, or any liquid, gaseous, or solid substance into water which alters the physical, chemical, or biological properties of water to make it harmful or injurious to public health and the health of animals or of aquatic organisms’.

This Act led to the creation of Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs). CPCB primarily promotes cleanliness of streams and wells in the different states of India.

The other functions of the board are:

1. To lay down the standards for a stream or well, and in consultation with the State Government concerned to modify or annul those standards.
2. To plan and cause the execution of a nationwide programme for the prevention, control, and abatement of water pollution.
3. To prepare manuals, codes, or guides related to the treatment and disposal of sewage and trade effluents. Also, collection, compilation, publication of technical and statistical data related to water pollution, and the measures devised for its effective prevention and control.
4. To advise the central government on any matter concerning the prevention and control of water pollution.
5. To provide technical assistance and guidance to the SPCBs and coordinate their activities.



Fig.2

6. To control the abatement of water pollution through investigation and research related to problems related to water pollution.
7. The functions of State Pollution Control Boards are similar to those of Central Pollution Control Board but they are to be executed at state level, and these are governed by the directions of CPCB.
8. The board advises the state government the location of any industry that might pollute a stream or a well.
9. The board is authorized to take samples from industrial streams, well or trade effluent or sewage, and it lays down the standards for effluents.
10. The board is licensed to take legal samples of trade effluents in accordance with the prescriptions in the act. In the presence of the occupier or his agent, the collected sample is divided into two parts, sealed and signed by both parties. Further, it is sent to some recognized lab for analysis. If the samples cross maximum permissible limits, then consent is refused to the unit.
11. Industries have to apply for consent from the board in a prescribed format, providing all technical details along with a prescribed fee. After application analysis of the effluent is carried out, then consent is provided for a fixed duration.
12. The board suggests efficient methods for utilisation, treatment, and disposal of trade effluents.



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15.5 FOREST (CONSERVATION) ACT, 1980

This Act was passed to prevent deforestation, which results in ecological imbalance and environmental deterioration. It prevents even the State Governments and any other authority dereserve a forest which is already reserved. It prohibits forestland to be used for non-forest purposes, except with the prior approval of the Central Government. The Salient features of Forest Act are:

1. The state government can use the forests only for forestry purposes. For non-forest purposes like mining or for monoculture of the economically important trees the state government has to take prior approval of central government.
2. Provisions for conservation of all types of forests and for this purpose there is an advisory committee which recommends funding for it to the central government.
3. Immediate halt to any illegal forest activity within a forest area.

1992 Amendment in the Forest Act

In 1992, a few amendments were made in the Act which made arrangements for permitting some non-forest exercises in forests, without cutting trees or restricted cutting with advance endorsement of central government. These activities involve setting of transmission lines, seismic reviews, investigation, drilling, and hydroelectric tasks. The last action includes large-scale destruction of forests, for which earlier endorsement by the central government is necessary.

1. Under this Act, investigation or review in wildlife sanctuaries, national parks, etc. is absolutely prohibited without prior endorsement by central government, even if no tree-felling is included.
2. Cultivation of cash crops like tea, coffee, spices, rubber, and plants are included as non-forestry activity and not permitted in reserve forests.
3. The central government has to endorse even the cultivation of fruit-bearing trees, oil-yielding plants, or plants of medicinal value in the forest area.



This is on the grounds that recently presented species in the forest area may cause an imbalance in the forest ecology. In case the species to be planted is a local species, then no prior clearance is required.

4. Tusser cultivation (a kind of silk-yielding insect) in forest areas by tribals is a method for their livelihood. It is treated as a forestry activity as long as the host trees are not some specific trees like Asan or Arjun. This is done so as to dishearten monoculture rehearses in the woodland which are otherwise wealthy in biodiversity.
5. Plantation of mulberry for rearing silkworm is viewed as a non-forest exercise. The explanation is same as described previously.
6. Mining is a non-forestry activity and prior endorsement of central government is required. For a case involving T.N. Godavarman Thirumulkpad vs. Association of India (1997), the Supreme Court guided all on-going mining exercises to be stopped quickly in any forest area of India if it had not been endorsed in advance by the central government.
7. Removal of stones, bajri, rock, and so on from riverbeds situated inside the forest zone falls under non-forest activity.
8. Any proposition sent to the central government for non-forest action must have a cost benefit analysis and Environmental Impact Statement (EIS) of the proposed exercise regarding its ecological and socio-economic effects.

Consequently, the Forests (Conservation) Act has made abundant arrangements for the preservation and security of forests and prevention of deforestation.

15.6 THE AIR (PREVENTION AND CONTROL OF POLLUTION) ACT, 1981

The Central Government used *Article 253* to enact this law and made it applicable throughout India. This Act defines air pollutant as ‘any solid, liquid, or gaseous substance (including noise) present in the atmosphere in such concentration as may be or tend to be injurious to human beings or other living creatures or plants or property or environment’. The CPCB and the SPCBs created under the



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Water Act 1974 are entrusted with the implementation of the provisions of the Act. The CPCB may:

1. Counsel the central government on any matter related to the improvement of air quality and prevention, control, or abatement of air pollution.
2. Plan and cause to execute a nation-wide programme for the prevention, control, or abatement of air pollution.
3. Provide technical assistance and guidance to the SPCBs and coordinate their activities.
4. Prepare manuals, codes or guides related to the prevention, control or abatement of air pollution and collect, compile, and publish technical as well as statistical data related to air pollution and the measures devised for its effective prevention, control, or abatement.
5. Lay down standards for the quality of air. The functions of the SPCBs also include inspection of any control equipment, industrial plant or manufacturing process, and to take steps for the prevention, control, or abatement of air pollution. The units identified as polluting industries should obtain approval before their establishment or/and continuing their operations. The SPCBs, in consultation with the state governments, wherever necessary, can exercise the following powers.
6. Declare certain areas within the state as air pollution control area and prohibit the use of certain fuels or appliances in that control area.
7. Set standards for the emission from automobiles and disallow discharge of the emission of any air pollutant beyond the permissible limit.
8. Approach court for restraining persons from causing air pollution and restrict certain industrial plants
9. Form authority to enter and inspect the premises of the polluters to take samples of air.



15.7 THE ENVIRONMENT (PROTECTION) ACT 1986

This act defines environment as ‘water, air and land and the interrelationship which exists among and between water, air and land, and human beings, other living creatures, plants, micro- organism and property’. It defines hazardous substance as ‘any substance or preparation which, by reasons of its chemical or physiochemical properties or handling, is liable to cause harm to human beings, other living creatures, plants, micro-organism, property or the environment.



Fig 3

This Act empowers the Central Government to:

1. Coordinate actions of the state governments, officers, and other authorities under the Act or any other law which is relatable to the objects of the Act.
2. Plan and execute a nation-wide programme for the prevention, control, and abatement of environmental pollution.
3. Lay down standards for the emission or discharge of environmental pollutants from various sources and setting maximum permissible limits.
4. Identify areas in which any industry, operations or processes, or class of industries, operations, or processes shall not be carried out.
5. Prevent accidents mediateding environmental pollution by laying down safeguards and setting procedures for remedial measures after such accidents.

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6. Identify materials, substances, and manufacturing processes, which can cause environmental pollution.
7. Carry out and sponsoring investigations and researches related to the problems of environmental pollution.
8. Inspect for any premises, plant, equipment, machinery, manufacturing or other processes, materials or substances as a source of pollution, and giving order to authorities to take steps for the prevention, control, and abatement of environmental pollution.
9. Establish or recognise environmental laboratories and institutions.
10. Prepare manuals, codes, or guides relating to the prevention, control and abatement of environmental pollution, and publication of information related to environmental pollution. The Central Government may organise an authority or authorities to make rules covering the following matters.
11. The permissible standards of air, water or soil quality for various areas and purposes.
12. The permissible limits of various environmental pollutants (including noise) for different areas.
13. The procedures, safety methods, prohibitions, and restrictions for the handling of hazardous substances in different areas

15.8 SUMMARY

The Constitution of India clearly states that it is the obligation of the state to ‘protect and improve the environment and to safeguard the forests and wildlife of the country’. It’s an obligation of every citizen ‘to ensure and improve the natural environment including forests, lakes, streams, and wildlife’. Directive Principles of State Policy as well as the Fundamental Rights has also referred to the environment. The established arrangements are upheld by various laws – acts, rules, and notices. The EPA (Environment Protection Act), 1986 came into power soon after the Bhopal Gas Tragedy and is viewed as an umbrella enactment as it fills numerous holes in the current laws. Hereon, an enormous number of laws



Environmental Law & Environmental Legislations in India

appeared as the issues started emerging. For instance, handling and management of hazardous waste rules in 1989.

The following environmental legislations have come into effect:

- **1972:** The Wildlife Protection Act, protects the birds and animals for all issues that are associated with them whether it be their habitat or the forests that provide for them.
- **1974:** The Water (Prevention and Control of Pollution) Act builds up an institutional structure for prevention and decrease in water contamination. It sets up the standards for water quality and effluents. Industries must look for permission to release waste into the water bodies.
- **1980:** The Forest (Conservation) Act and Rules, 1981, accommodates the conservation and protection of the forests.
- **1981:** The Air (Prevention and Control of Pollution) Act accommodates the control and reduction of air contamination. It provides CPCB the authority for successful implementation of this act.
- **1986:** The Environment (Protection) Act enables the central government to secure and improve ecological quality, control and diminish contamination from all sources, and forbid or limit the setting and/or activity of industries on environmental grounds. The CPCB (Central Pollution Control Board) was comprised under this demonstration.
- **2006:** The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, was implemented to protect the forest rights and occupation of forest land in forest dwelling tribes, in order to balance their socio-economic class and their right to environment.

15.9 SELF-ASSESSMENT QUESTIONS

1. Describe about the environmental protection act, 1986.
2. Briefly describe about the water act, 1972.
3. What do you understand by Environmental legislation and how this can be beneficial for environmental protection?

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LESSON 16

HUMAN COMMUNITIES AND THE ENVIRONMENT

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Structure

- 16.1 Learning objective
- 16.2 Introduction
- 16.3 Factors Responsible for Human Population Growth
- 16.4 Impacts of Human Population Growth on Environment, Human Health, and Welfare
- 16.5 Steps for Population Control
- 16.6 Carbon Footprint
- 16.7 Resettlement and Rehabilitation of Developmental Project Affected Persons and Communities
- 16.8 CNG (Compressed Natural Gas) Vehicles Initiative in Delhi
- 16.9 Summary
- 16.10 Self-Assessment Questions
- 16.11 References

16.1 LEARNING OBJECTIVES

Gain understanding about the various factors responsible for the human population explosion, the negative impacts of the ever-growing human population on the environment and human health.

1. Develop knowledge about certain measures and policy decisions that can be adopted to keep a check on the growing population.
2. Examine the challenges associated with Resettlement and rehabilitation of project affected families and communities, outline the legal framework available to formulate an effective plan for the resettlement of the displaced people.

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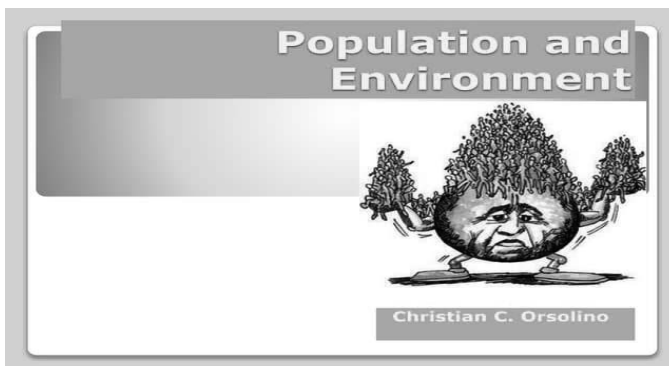


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3. Understand the importance of communication for protection of environment and the means to disseminate information.
4. Gain understanding about the various environmental movements that have taken place in India and that have helped in progress of environmentalism in the country.
5. Be aware about some initiatives by Ministry of Environment, Forests, and Climate Change.

16.2 INTRODUCTION

Human population growth is the genesis of most of the environmental issues we face today and can be linked with negative environmental, economic and social impacts. The environmental impacts may include deforestation, land degradation, land-use change, water crisis, and climate change. Several steps have been taken to ensure sustainability in growth and development, but the ever-growing human population keeps on challenging these initiatives.



The current human population on earth is over 7.8 billion and is increasing with projections of over 9 billion by 2040 and 11 billion by 2100. The rapidly growing population is far outpacing the carrying capacity of the planet, which is the ability of the planet to support the overly sized population of humans as per the available resources. Overpopulation is the genesis of most of the environmental issues we face today and can be linked with negative environmental, economic and social impacts.



16.3 FACTORS RESPONSIBLE FOR HUMAN POPULATION GROWTH

Growth in the human population can be largely attributed to some advancement in the recent past in the field of agriculture, science, technology, and health facilities. The surplus of food and other supplementary resources has provided ample space for the growing human population. Overpopulation at specific locations, mainly cities can also result from migration in search of better living conditions. Several factors that lead to such unprecedented growth in the human population are:

- 1. Increase in agricultural productivity:** Inclusion of modern techniques in agriculture like use of agrochemicals (like fertilizers, pesticides, and insecticides) and modern machinery has led to multi-fold increase in the agricultural yield. The surplus of food resources have been able to support the growing human population.
- 2. Decrease in death rate and infant mortality rate:** With the advancement in medical facilities including improved medicines and vaccines there has been a considerable worldwide decline in the death rate and infant mortality rate, which has led to larger population size.
- 3. Advancement in Science and Technology:** Advance scientific technology which fueled the industrial revolution provided a surplus of resources that could support a larger human population.
- 4. Lack of awareness:** Illiteracy, poverty, and lack of awareness about the harmful impacts of a large population have also led to the problem of overpopulation. Awareness and use of various family planning methods are also lacking in most highly populated countries.

16.4 IMPACTS OF HUMAN POPULATION GROWTH ON ENVIRONMENT, HUMAN HEALTH, AND WELFARE

Following are some of the major environmental and health impacts of growing human population:



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1. **Deforestation:** The growing human population will lead to an increased demand for food, space, and energy which will ultimately lead to the destruction and clearing of pristine forest areas. In order to increase the agricultural yield to support a large population, new areas have to be brought under agriculture which will lead to a large scale deforestation. Forests will also be cleared due to urbanisation, infrastructure development, and industrialisation. Deforestation will lead to reduced carbon dioxide absorption which will ultimately lead to global climate change phenomenon.
2. **Biodiversity loss:** Decline in forest cover and other available habitats will also lead to a decline in floral and faunal biodiversity. This will lead to an ecological imbalance and a decline in the biological resource of the planet.
3. **Urbanisation:** Rapidly growing human population has led to rapid urbanisation which has adversely affected the environment and natural resources. Due to the growing population, natural resources in the urban areas are depleting at a faster rate. A large part of the urban population is devoid of proper sanitation facilities and pure drinking water which has resulted in adverse health impacts.
4. **Industrialisation:** The establishment of industries such as fertilizers, cement, steel, chemicals, and oil refineries to support the needs of the growing population leads to environmental degradation, land-use change, and pollution. Growing levels of toxic pollutants in the atmosphere have a detrimental effect on the respiratory health of the affected people.
5. **Land Degradation:** Intensive farming, excessive use of agrochemicals, and overexploitation of land and water resources leads to soil degradation, including soil erosion, salination, and desertification.
6. **Generation of Waste:** The large human population generates humungous solid and liquid waste which has detrimental environmental and health impacts.
7. **Loss of Fresh Water:** Overexploitation of both surface and groundwater leads to a decline in the available fresh water resources. It leads to water



scarcity issues and unavailability of safe drinking water. It is projected that 2/3rd of the world's population will be living with water shortages by 2025.

8. **Climate Change:** Overpopulation and excessive use of fossil fuels have increased the level of various gaseous pollutants in the atmosphere including greenhouse gases.
9. **Transport Infrastructure:** Development of transport infrastructure for the large human population has led to the release of large quantities of poisonous gases such as carbon monoxide, nitrogen oxides, and hydrocarbons. The development of ports and harbours in the sea has led to oil spills from ships which adversely affect the marine biodiversity and ecosystems.
10. **Productivity:** Environmental degradation not only harms the environment but also reduces economic productivity. Polluted air and water causes several diseases in different parts of the world and thus reduces the productivity of the country. Water pollution has led to a decline in fisheries and water availability for agriculture. Soil degradation leads to the siltation of reservoirs and canals and transport channels.

16.5 STEPS FOR POPULATION CONTROL

Health and Welfare

India has the largest population (1.3 billion) in the world and it is still growing. Hence, effective population control measures have become the need of the hour. Following measures can be taken to control the population growth in India:

1. **Minimum age of Marriage:** In India, the minimum age for marriage is 21 years for men and 18 years for women. This law should be effectively implemented and people should be made aware regarding the same through publicity.

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2. **Better Education:** Education to different strata of society changes the outlook of people. Educated families prefer to delay marriage and adopt small family norms.
3. **More employment opportunities:** Developing employment opportunities in both urban and rural areas will improve people's income and standard of living and thus their chances of adopting small family norms.
4. **Family Planning:** Increasing awareness about various family planning methods and their easy availability can ensure their use and thus help in reducing family size.
5. **Incentives:** The government can provide various types of incentives to the people to adopt birth control measures. This can include monetary incentives, tax benefits, and other facilities like leave and promotion which can be extended to the working class which adopts small family norms.

16.6 CARBON FOOTPRINT

Carbon Footprint is the amount of greenhouse gases mainly carbon dioxide released into the atmosphere by any anthropogenic activity. It can be a broad measurement of the environmental impact due to the actions of an individual, family, organization, industry, or even an entire nation. Example: To calculate an individual's carbon footprint we need to include a rough estimate of our overall carbon dioxide production from several daily activities including the use of electrical appliances, the distance we drive, our food consumption, and how much we recycle.



Our carbon footprint can be reduced by using environment-friendly vehicles (battery operated), using public transportation, using energy efficient appliances, insulating our homes to reduce electricity consumption on room heaters and air conditioning, and consuming local and organic food. Individuals and companies can also offset some of their carbon dioxide emissions by purchasing carbon credits, the money from which can go into environment-friendly initiatives such as planting trees or investing in renewable and clean energy.

16.7 RESETTLEMENT AND REHABILITATION OF DEVELOPMENTAL PROJECT AFFECTED PERSONS AND COMMUNITIES

Displacement of human populations in India is mostly initiated by factors such as large scale developmental projects like mining, dam building, and industrial establishments apart from political conflicts, natural disasters, and declaration of an area as a National Park, Wildlife Sanctuary, or Biosphere Reserve. In India, a national policy on Resettlement and Rehabilitation of project-affected families came into force in 2004. However, all displacement and resettlement programs have faced several challenges due to regional issues and administrative delays.

Resettlement and Rehabilitation (R&R) of families and communities displaced during the establishment of any large-scale developmental project is an integral part of the Environmental Impact Assessment and Environmental Management Plan Reports (EIA and EMP). The R&R Plan for project-affected families for any proposed or ongoing is prepared based on the National Policy for Rehabilitation and Resettlement, 2003, and as per National Rehabilitation and Resettlement Policy, 2007 (NRRP-2007). The resettlement plan is implemented in association with the concerned state government, representatives from the project affected families, and other stakeholders in the area. A detailed socio-economic survey has to be conducted before the formulation of the R&R Plan. Large-scale infrastructure projects involve the involuntary displacement of many people which gives rise to severe economic, social, and environmental problems. People are relocated to new places where their productive skills may be less applicable, production systems are degraded, employment opportunities



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and income sources are lost. The social fabric within the community is broken if the families from one area are displaced to separate locations. The resettlement of affected people (oustees) must aim to minimize involuntary resettlement, provide a time-bound resettlement plan, and compensation for land and other assets affected by the project.

16.8 CNG (COMPRESSED NATURAL GAS) VEHICLES INITIATIVE IN DELHI

Rising levels of pollution have been a major cause of concern in the National Capital, Delhi, which has been put on the list of 20 most polluted cities in the world by the World Health Organisation. The main issues related to environmental pollution are air pollution caused due to vehicular and industrial pollution and water pollution in the Yamuna River. The number of private and public vehicles has increased many folds in the recent past. This has polluted the environment and made people, especially children and old vulnerable to multiple diseases in the city.

Reacting to the court order which was the result of a PIL, the government made it compulsory to introduce the CNG vehicles and make the pollution check mandatory for all private vehicles. The introduction of the CNG vehicles has resulted in the reduction of the environmental pollution in the city. Similarly, the Delhi government has been forced to shift the polluting industries out of the city and launch the Yamuna River cleaning operation. Moreover, taking action as per recent orders of the Supreme Court of India to check it, the Delhi Government is planning to push hydrogen enriched CNG buses on its roads. From just five lakh vehicles in 1981, the increase in the number of vehicles has tremendously proliferated. According to Delhi traffic police, around eighty thousand vehicles travel through Delhi every day between 8pm and 6am indicating the size of its fleet. On 28 July 1998, the Supreme Court ordered all commercial vehicles to be mandatorily converted to CNG vehicles, especially taxis, buses, and three-wheelers.

Advantages of using CNG: CNG may be generated and used for bulk storage and pipeline transport of renewable energy and also be mixed with biomethane,



itself derived from biogas from landfills or anaerobic digestion. This would allow the use of CNG for mobility without increasing the concentration of carbon in the atmosphere. It would also allow continued use of CNG vehicles currently powered by non-renewable fossil fuels that do not become obsolete when stricter CO₂ emissions regulations are mandated to combat global warming.

Limitations of using CNG: The use of CNG vehicles faces several limitations such as fuel storage and infrastructure available for delivery and distribution at fuelling stations. CNG must be stored in high pressure cylinders only and this cylinder takes up more space than gasoline or diesel tanks. Other limitation include relatively price and environmental insensitivity.

Due to burning issues like climate change and estimations that non-renewable fuels like crude oil will get exhausted in near future, exploring new fuels like CNG is a boon. Thus the government shall now take more initiatives so that alternative fuels can be commercialised.

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16.9 SUMMARY

1. Challenges associated with resettlement and rehabilitation of project affected families and communities.
2. Legal frameworks are available to formulate effective resettlement plan.
3. Negative impacts of the ever-growing human population on the environment and human health.
4. Factors responsible for the human population explosion.
5. Population control measures.

16.10 SELF-ASSESSMENT QUESTIONS

1. What is environmental movement ? Do you think that socio-cultural factors play a crucial role in the movement processes which are taking place in rural India?
2. What is Chipko Movement ?



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3. What is environmental movement? Do you think that India has enough laws to protect the natural environment of the country ?

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LESSON 17

ENVIRONMENTAL MOVEMENTS IN INDIA

NOTES

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Structure

- 17.1 Learning Objective
- 17.2 Introduction
- 17.3 Bishnoi Movement
- 17.4 Chipko Movement
- 17.5 Silent Valley Movement
- 17.6 Appiko Movement
- 17.7 Narmada Bachao Andolan
- 17.8 Summary
- 17.9 Self-Assessment Questions
- 17.10 References

17.1 LEARNING OBJECTIVES

In this unit, our focus will be on environmental and ecological movements. While agrarian or working class movements have had a long historical trajectory, environmental, or ecological movements gained worldwide attention only in the second half of the twentieth century. These movements focus not only on basic survival issues but also on larger ecological concerns. These are different from earlier social movements and there is a need to understand them in terms of their nature and strategies.

17.2 INTRODUCTION

Environmental and ecological movements are among the important examples of the collective actions of several social groups. Protection and recognition of constitutional and democratic rights, which are not defined by law but form an

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important part of the day to day living of the subaltern masses like the control over their resources, the right of indigenous people to preserve their culture, protection of environment, and maintenance of ecological balance are significant concerns of these movements as they affect the human life to a great extent. These movements also reflect an enlarged vision of economics and politics. Economic justice sought by these movements does not mean mere distribution of resources but encompass a larger vision like enhancement in the quality of life through recognition of people's right over their natural resources, their right to live with dignity, and their participation in the decision making. The concerns of human environment received spectacular attention of scholars following the conclusion of the United Nations Conference on Human Environment, Stockholm in 1972. By the 1980s the "green movement" became a worldwide phenomenon encompassing various countries of the world including India. It is signified by several movements of people for the protection of their environmental and ecological rights in India, 'eco-greens' or 'green movement' in Germany and North America.

17.3 BISHNOI MOVEMENT



Fig. 1

This was begun 400 years ago by a Sage known as Sombaji. In Rajasthan, a large number of trees are still worshiped by devotees. People resisted the cutting of such trees and advocated movement against deforestation.



Environmental Movements in India

- **Year:** 1700s
- **Place:** Khejarli, Marwar region, Rajasthan
- **Leaders:** Amrita Devi along with Bishnoi villagers in Khejarli and surrounding villages.
- **Aim:** Saving sacred trees from being cut down by the king's soldiers for a new palace.

Amrita Devi, a female villager could not bear to witness the destruction of both her faith and the village's sacred trees. She hugged the trees and encouraged others to do the same. Totally 363 Bishnoi villagers were killed in this movement. The Bishnoi tree martyrs were influenced by the teachings of *Guru Maharaj Jambaji*, who founded the Bishnoi faith in 1485 and set forth principles forbidding harm to trees and animals. The king who came to know about these events rushed to the village and apologized, ordering the soldiers to cease logging operations. Soon afterwards, the maharaja designated the Bishnoi state as a protected area, forbidding harm to trees and animals. This legislation still exists today in the region.

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17.4 CHIPKO MOVEMENT



Fig. 2



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The Chipko movement of Uttaranchal is famous for its tree-hugging campaign to resist the tree cutting. It was started by noted environmentalist Sunderlal Bahuguna in 1970 to safe guard the rich forest of western Himalaya Range.

- **Year:** 1973
- **Place:** In Chamoli district and later at Tehri-Garhwal district of Uttarakhand.
- **Leaders:** Sundarlal Bahuguna, Gaura Devi, Sudesha Devi, Bachni Devi, Chandi Prasad Bhatt, Govind Singh Rawat, Dhoom Singh Negi, Shamsheer Singh Bisht, and Ghanasyam Raturi.
- **Aim:** The main objective was to protect the trees on the Himalayan slopes from the axes of contractors of the forest.

Mr. Bahuguna enlightened the villagers by conveying the importance of trees in the environment which checks the erosion of soil, cause rains, and provides pure air. The women of Advani village of Tehri-Garhwal tied the sacred thread around trunks of trees and they hugged the trees, hence it was called ‘*Chipko Movement*’ or ‘*hug the tree movement*’. The main demand of the people in these protests was that the benefits of the forests (especially the right to fodder) should go to local people. The Chipko movement gathered momentum in 1978 when the women faced police firings and other tortures. The then state Chief Minister, Hemwati Nandan Bahuguna set up a committee to look into the matter, which eventually ruled in favour of the villagers. This became a turning point in the history of eco-development struggles in the region and around the world.

17.5 SAVE SILENT VALLEY MOVEMENT

Save Silent Valley was an environmental movement intended to protect Silent Valley, an evergreen tropical forest in the Palakkad district of Kerala, India. It was started through a hydroelectric project.

- **Year:** 1978
- **Place:** Silent Valley, an evergreen tropical forest in the Palakkad district of Kerala, India.



- **Leaders:** The Kerala Sastra Sahitya Parishad (KSSP), an NGO, and the poet-activist Sughathakumari played an important role in the Silent Valley protests.
- **Aim:** In order to protect the Silent Valley, the moist evergreen forest from being destroyed by a hydroelectric project.

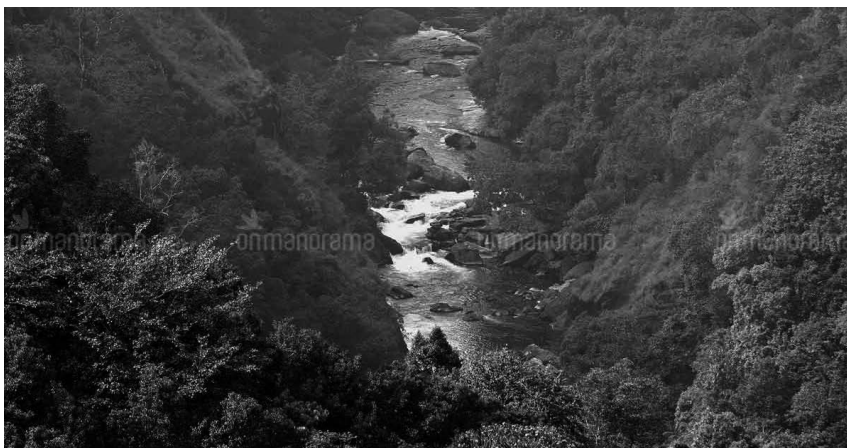


Fig. 3

The Kerala State Electricity Board (**KSEB**) proposed a hydroelectric dam across the Kunthipuzha River that runs through the Silent Valley. In February 1973, the planning commission approved the project at a cost of about Rs. 25 crores. Many feared that the project would submerge 8.3 sq km of untouched moist evergreen forest. Several NGOs strongly opposed the project and urged the government to abandon it. In January 1981, bowing to unrelenting public pressure, Indira Gandhi declared that Silent Valley will be protected. In June 1983, the center re-examined the issue through a commission chaired by Prof. M.G.K. Menon. In November 1983, the Silent Valley Hydroelectric Project was called off. In 1985, Prime Minister Rajiv Gandhi formally inaugurated the Silent Valley National Park.

17.6 APPIKO MOVEMENT

The Appiko movement was an innovative movement based on the environmental conservation in India. This movement focused on the issues of forest based environmental actions in India.

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- **Year:** 1983
- **Place:** Uttara Kannada and Shimoga districts of Karnataka State
- **Leaders:** Appiko's greatest strengths lie in it being neither driven by a personality nor having been formally institutionalised. However, it does have a facilitator in Pandurang Hegde. He helped launch the movement in 1983.
- **Aim:** Against the felling and commercialisation of natural forest and the ruining of ancient livelihood.



Fig. 4

It can be said that Appiko movement is the southern version of the Chipko movement. The Appiko Movement was locally known as “Appiko Chaluvali”. The locals embraced the trees which were to be cut by contractors of the forest department. The Appiko movement used various techniques to raise awareness such as foot marches in the interior forest, slide shows, folk dances, street plays, etc. The second area of the movement's work was to promote afforestation on denuded lands. The movement later focused on the rational use of ecosphere by introducing alternative energy resource to reduce the pressure on forest. The movement became a success. The current status of the project is stopped.



17.7 NARMADA BACHAO ANDHOLAN (NBA)

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The Narmada movement begun in the late 1970s. This movement was centred on the issue of human rights and rehabilitation programmes for the dam displaced.

- **Year:** 1985
- **Place:** Narmada River, which flows through the states of Gujarat, Madhya Pradesh, and Maharashtra.
- **Leaders:** Medha Patker, Baba Amte, adivasis, farmers, environmentalists, and human rights activists.
- **Aim:** A social movement against a number of large dams being built across the Narmada River.



Fig. 5

The movement first started as a protest for not providing proper rehabilitation and resettlement for the people who have been displaced by the construction of Sardar Sarovar Dam. Later on, the movement turned its focus on the preservation of the environment and the eco-systems of the valley. Activists also demanded the height of the dam to be reduced to 88m from the proposed height of 130m. World Bank withdrew from the project. The environmental issue was taken into the court. In October 2000, the Supreme Court gave a judgment approving the construction of the Sardar Sarovar Dam with a condition that height of the dam could be raised to 90 M. This height is much higher than the 88 M which anti-dam activists demanded, but it is definitely lower than the proposed



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height of 130 M. The project is now largely financed by the state governments and market borrowings. The project is expected to be fully completed by 2025.

17.8 SUMMARY

1. The environmental movement also including conservation and green politics is a dissimilar scientific, social, and political movement to address environmental issues. Environmental movement is a type of “social movement that involves an array of persons, groups and coalitions that observe a common interest in environmental protection and act to bring about changes in environmental policies and practices” (Tong, Yanki 2005).
2. The environmental movements favour the sustainable management of natural resources. The movements often stress the protection of the environment via changes in public policy. Many movements are centred on ecology, health and human rights.
3. Environmental movements range from the highly organised and formally institutionalised ones to the radically informal activities.
4. The spatial scope of various environmental movements ranges from being local to the almost global.

17.9 SELF-ASSESSMENT QUESTIONS

1. Highlight the main issues and concerns of the environmental movements in India.
2. Discuss the main feature of the Chipko movement.
3. Write a note on Narmada Bachao Andolan (NBA).
4. In your opinion, how are the environmental and ecological rights related to democracy and development in India? Explain.



NOTES

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LESSON 18

DISASTER MANAGEMENT

NOTES

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Structure

- 18.1 Learning Objectives
- 18.2 Introduction
- 18.3 ‘Post Disaster Management’ and ‘Disaster Mitigation
- 18.4 Multi-Disciplinary and Multi- Sectoral nature of Disaster Management:
- 18.5 Guidelines for effective management of mitigation program
- 18.6 The main elements of a mitigation strategy
- 18.7 Causes, Effects and Mitigation measures of the disasters in India:
 - 18.7.1 Flood
 - 18.7.2 Earthquakes
 - 18.7.3 Cyclones
 - 18.7.4 Landslides:
 - 18.7.5 Tsunami
- 18.8 Summary
- 18.9 Internal Assessment Questions
- 18.10 References

18.1 LEARNING OBJECTIVES

India, with its vast territory, large population, and unique geo-climatic conditions has experienced many extreme hazardous events that have turned into disasters in the last few decades. Floods, droughts, cyclones, earthquakes, and landslides are recurrent phenomena in the country. Susceptibility to such extreme natural events is compounded by frequent occurrences of human made disasters such as fire, epidemics, stampedes, chemical leakages, etc. This Unit aims at providing you with a comprehensive understanding of terminology and concepts used in the management of disasters and their relationship with each other. You will also learn about the linkage of the process of disaster management with developmental planning. After reading this lesson, you should be able to:

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- I. Explain the meaning of hazard, risk, vulnerability, and capacity with respect to disaster management.
- II. Describe the relationship between hazard, vulnerability, risk, and capacity.
- III. Discuss various models of Disaster Management.
- IV. Illustrate the tools used in hazard, vulnerability, risk, and capacity analysis.

18.2 INTRODUCTION

A disaster is a serious problem occurring over a period of time that causes widespread human, material, economic, or environmental loss which exceeds the ability of the affected community or society to cope using its own resources. The Indian subcontinent is very vulnerable to droughts, floods, cyclones, earthquakes, landslides, avalanches, and forest fires. Disasters are routinely divided into either “natural disasters” caused by natural hazards or “human-instigated disasters” caused from anthropogenic hazards. However, in modern times, the divide between natural, human-made, and human-accelerated disasters is difficult to draw. Disaster Management as a subject essentially deals with management of resources and information as far as a disastrous event is concerned and also how effectively and seamlessly one coordinates these resources.

Disaster management, at the individual and organisational level, deals with issues of planning, coordinating, communication and risk management. A book to teach disaster management would accomplish very little unless a broad spectrum of knowledge is imparted regarding effects of various types of disasters, their dimensions, and characteristics, the role of United Nations in complex emergencies and the role of Indian government in prevention and control of disasters. Equally important is the knowledge about various international and national agencies involved in disaster relief and humanitarian assistance. This book covers all these subjects in order to enhance the ability of students and professionals for better disaster response.



18.3 ‘POST DISASTER MANAGEMENT’ AND ‘DISASTER MITIGATION’

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Disaster management is a process of effectively preparing for and responding to disasters. It involves strategically organizing resources to lessen the harm that disasters cause. It also involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response, and recovery. It is multi-sectoral as it involves administrators, scientists, planners, volunteers, and communities. These roles and activities span the pre-disaster, during disaster, and post disaster plans. Since their activities are complementary as well as supplementary to each other, there is a critical need for coordinating these activities. The Indian subcontinent is very vulnerable to droughts, floods, cyclones, earthquakes, landslides, and forest fires. While not all natural calamities can be predicted and prevented, a state of preparedness and ability to respond quickly to natural calamity can considerably mitigate loss of life, property, human suffering, and restore normalcy at the earliest.

The post disaster approach towards developing a set of strategies to assist a community in rebuilding after a disaster occurs. It deals with natural disasters, involving problems such as evacuation, warnings, communications, search and rescue, fire-fighting, medical and psychiatric assistance, provision of relief, shelter, etc, is generally referred to as ‘Post Disaster Management’. It is a primarily a ‘Reactive Mechanism’ to the natural disasters.



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Today we have a range of early warning systems for a range of natural hazards. Although they are more accurate than before and can help in prediction but it is not enough to ensure communities are safe from disasters. This is where disaster mitigation can play an important role. ‘*Mitigation*’ is defined as a sustained action taken to reduce long term vulnerability of human life and the recovery property to natural hazards while the preparatory, response, and the recovery phases of emergency management relate to specific events. Mitigation activities have the potential to produce repetitive benefits over time. Though traditionally disaster management consisted primarily of reactive mechanisms, the past few years have witnessed a gradual shift towards a more proactive, mitigation based approach i.e. a ‘*Proactive approach*’ to natural disasters.

Goals - The goals of post-disaster recovery planning are to:

1. **To identify and prioritize key issues.**
2. **Establish partnerships.**
3. **Develop a recovery strategy.**
4. **Effectively direct external and internal resources.**
5. **Identify per-disaster mitigation projects.**
6. **Enhance response and preparedness capabilities.**

Disaster Preparedness: Preparedness focuses on plans to respond to a disaster threat or occurrence. It takes into account an estimation of emergency needs and identifies the resources to meet these needs. It also involves preparation of well-designed plans to structure the entire post-disaster response, and familiarising the stakeholders, particularly the communities through training and simulation exercises. Preparedness has to be supported by the necessary legislation, means a readiness to cope with disasters or similar emergencies that cannot be avoided. The first objective of preparedness is to reduce the disaster impact through appropriate actions and improve the capacity of those who are likely to be affected most (that is, marginalised, poor and handicapped) to get maximum benefit out of relief. The second is to ensure that ongoing development continues to improve the capacities and the capabilities of the system to strengthen preparedness and efforts at community level. Finally, it guides reconstruction so as to ensure reduction in vulnerability. The best examples of preparedness activities are the development of local warning and community evacuation plans through community education, evolving local response structures such as Community based Disaster



Management Teams (DMT) and administrative preparedness by way of stockpiling of supplies; developing emergency plans for rescue and relief.

18.4 MULTI-DISCIPLINARY AND MULTI- SECTORAL NATURE OF DISASTER MANAGEMENT

Disaster management is a multidisciplinary area in which a wide range of issues that range from forecasting, warning, evacuation, search and rescue, relief, reconstruction, and rehabilitation are included. It is also multi-sectoral as it involves administrators, scientists, planners, volunteers, and communities.

India is a parliamentary democracy with a federal structure. An integrated disaster management mechanism exists within government framework. The essential responsibility of disaster management lies with the State Government where the disaster has occurred. However, in the event of disasters which spread over several states and assume uncontrollable proportions, the Central Government may be required to supplement appropriate measures in the fields of rescue, relief, and preparedness

18.5 GUIDELINES FOR EFFECTIVE MANAGEMENT OF MITIGATION PROGRAM

The National Disaster Management Authority (NDMA), headed by the Prime Minister of India, is the apex body for Disaster Management in India-

1. Pre-disaster mitigation can help in ensuring faster recovery from the impacts of disasters.
2. Mitigation measures must ensure protection of the natural and cultural assets of the community.
3. Hazard reduction methods must take into account the various hazards faced by the affected community, their desires, and priorities.
4. Any mitigation programme must also ensure an effective partnership between the Govt, Scientific, private sector, NGOs, and the community.



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18.6 THE MAIN ELEMENTS OF A MITIGATION STRATEGY



Fig 2

Disaster mitigation focuses on the hazard that causes the disaster and tries to eliminate or drastically reduce its direct effects. The best example of mitigation is the construction of dams or levees to prevent floods or coordination of release of water from various irrigation dams to avoid flooding in the downstream areas. Other examples include strengthening buildings to make them earthquake resistant, planting of crops that are less affected by disasters, controlling land-use patterns to restrict development in high-risk areas and diversification of economic activities to act as insurance to offset losses in different sectors. A mitigation strategy however, cannot be successful unless it has the backing and support of all concerned – the administrative machinery, the research institutions, the non-officials, and the community. So, it also becomes imperative to have built-in institutional arrangements and/or legislative backing to oversee the mitigation strategy over a period of time.

The main elements of mitigation strategy are-

- 1. Risk assessment and Vulnerability analysis:** This involves the identification of hotspot areas of prime concern, collection of information on past natural hazards, information on the population and infrastructure.



2. **Applied research and technology transfer:** There is a need to establish or upgrade observation, equipment and networks, monitor the hazardous properly, improve the quality of forecasting, and warning.
3. **Public awareness and training:** Training to be given to officials and staff of various departments involved in state & district level.
4. **Institutional mechanisms:** There is need to emphasize on proactive and pre-disaster measures rather than post-disaster response. It is thus essential to have a permanent administrative structure which can monitor the developmental activities across departments and provides suggestions for necessary mitigation measures. The national disaster management center (NDMC) can perform such a task. Professionals like architects, structural engineers, doctors and chemical engineers who are involved with management of hazardous chemicals, can be asked to form groups that can design specific mitigation measures.
5. **Incentives and resources for mitigation:** Provides stable source of funding for all mitigation programs.
6. **Land use planning and regulations:** Designate floodplains areas as places for outdoor recreation, wildlife attractions, or hiking trails. They can also urge people to avoid these areas during flood season.
7. **Hazard resistant design and construction:** In areas that are prone to disasters, protection can be enhanced by careful selection of sites and the way the buildings are built. Thus it is essential to promote the knowledge of disaster resistant construction techniques and practices among engineers, architects, and technical personnel.
8. **Structural and Constructional reinforcement of existing buildings:** It is also possible to reduce the vulnerability of existing buildings through minor adaptations or alterations thereby ensuring their safety. This can be done by the insertion of walls, especially on choked frames, construction of new frame systems, designing residential electrical equipment above flood level, designing water storage tanks to be able to withstand cyclonic winds, earthquakes, and floods.

In disaster situations, a quick rescue and relief mission is inevitable. However, considerable damage can be minimized if adequate preparedness levels



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are achieved. Indeed, it has been noticed in the past, that as and when attention has been paid to adequate preparedness measures, the loss to life and property has been considerably reduced. Preparedness measures, such as training of role players, including the community, development of advanced forecasting systems, effective communications, and above all, a sound national policy and a well-networked institutional structure involving government organisations, academic and research institutions, the armed forces and **NGOs** would greatly contribute to the overall disaster management of the region. Later, it would lead to the much needed change in the overall orientation from rescue and relief to preparedness.

18.7 CAUSES, EFFECTS AND MITIGATION MEASURES OF THE DISASTERS IN INDIA

18.7.1 Flood:

Based on hydrological studies carried out, it is estimated that only 18 percent of the rainwater can be stored in dams, reservoirs, etc. while 82 percent of the rainwater flows through rivers ultimately into the sea. Floods are therefore a recurring phenomenon in our country. Floods can be caused by natural, ecological, or anthropogenic factors either individually or as a combined result. Human activities such as deforestation and shifting cultivation can also contribute to floods. Heavy rainfall is the main cause of floods in the rivers. The breaches to tanks and reservoirs due to inflow of large quantities of water from excessive rainfall also result in floods. Floods occur sometimes in a flash due to intensive rains at the time of cyclones. India is the most affected country in the world. The west coast of India has an advantage, as it has Western Ghats with thick forests which acts as a natural buffer to floods.

Flooding in rivers is mainly caused by:

- Inadequate capacity within the banks of the river to contain high flows.
- River bank erosion and silting of riverbeds.



- Landslides leading to obstruction of flow and change in the river course.
- Synchronization of flood in the main and tributary rivers.
- Flow retardation due to tidal and backwater effects.
- Poor natural drainage.
- Cyclone and heavy rainfall.

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Effects:

Floods cause heavy suffering to people living in low lying areas because the houses and the properties are inundated or washed away. Most of the victims are rural folks who are economically poor. Floods also damage standing crops and livestock. Mitigation measures includes structural measures:

1. Reservoirs for impounding monsoon flows to be released in a regular manner.
2. After the peak flood flow passes.
3. Prevention of overbank spilling by the construction of embankments and flood walls.
4. Improvement of flow conditions in the channel and anti-erosion measures.
5. Improved drainage.



Fig 3



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Flood-Plain management such as:

1. Flood Plain zoning and flood proofing including disaster preparedness.
2. Maintaining wet lands.
3. Flood forecasting and warning services.
4. Disaster relief and public health measures.
5. Flood insurance.

The various measures adopted for flood mitigation may be categorized into two groups:

- i) Structural
- ii) Non-structural

The general approach was aimed at preventing floodwaters from reaching the potential damage centres, as a result of which a large number of embankments came up along the various flood prone rivers. The main thrust of the flood protection programme undertaken in the country so far in the form of structural measures may be grouped into the following:

- Dams and Reservoirs
- Embankments, flood walls, and sea walls
- Natural detention basin
- Channel improvement
- Drainage improvement
- Diversion of floodwaters.

Post Flood Management :

Post – Disaster Management could inter alia including the following essential requirements:

- Speedy restoration of roads, rail routes, and the postal services.
- Normal water supply in the affected areas either by arranging tankers or fire tenders.
- Repair of the power, telephone, and sewerage lines on priority basis so that normalcy is restored.



- Proper arrangements for the supply of food, shelter, and clothing to the marooned people.
- Ensuring adequate supply of **POL** and kerosene oil and keeping the supply line moving.
- Constitution of a survey team to assess the loss and compensation to be given to the affected population.
- Assistance for repairing or rebuilding of private properties.
- Desilting and dewatering of the inundated areas.
- Taking up appropriate measures according to Contingency Plan for the agriculture sector.

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Arrangements During and After Floods:

- i) Organising rescue operations.
- ii) Organising shelter for the people in distress. In case the efforts of the civil authorities are considered inadequate, army assistance should be requisitioned.
- iii) Relief measures by non-official and voluntary organisations may be enlisted as far as possible.
- iv) Organise relief camps.
- v) Provision of basic amenities like drinking water, sanitation and public health care, and arrangements of cooked food in the relief camps.
- vi) Making necessary arrangements for air dropping of food packets in the marooned villages through helicopters.
- vii) Organising enough relief parties to the rescue of the marooned people within a reasonable time limit.
- viii) Establish alternate communication links to have effective communication with marooned areas.
- ix) Organising controlled kitchens to supply foods initially at least for 3 days.
- x) Organising cattle camps, if necessary, and provide veterinary care, fodder, and cattle feed to the affected animals.
- xi) Grant of emergency relief to all the affected people.



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- xii) Submission of daily reports and disseminate correct information through mass media to avoid rumours.
- xiii) Rehabilitation of homeless.
- xiv) Commencement of agricultural activities – desiltation and resowing.
- xv) Repairs and reconstruction of infrastructure facilities such as roads, embankments, and resettlement of flood prone areas.
- xvi) Health measures.
- xvii) Relief for economic reconstruction.

18.7.2 Earthquakes:

An abrupt and tremendous release of energy stored in the rocks and the earth's crust through the action of tectonic process is described as an earthquake. Some areas on the planet earth are so located that more than one tectonic plate constantly change against one another and an earthquake can result when one or more of these plates move against the others at high speed. In certain regions of earth, earthquakes occur with regularity. In India, Assam and the Himalayan regions are more infested with earthquakes. Earthquake that occurred in 2001 in Gujarat taking a toll of 30,000 per ape was the major one Hythi at present.

The intensity of earthquake is measured on the Richter scale. As of now there is no way of predicting an earthquake. Earthquake by themselves do not cause casualties but the houses collapse due to poor construction. The construction of quake-proof houses may reduce human loss but it may be economically difficult for poor countries. The Government of countries which are quake prone should ensure that the infrastructure could withstand earthquakes. The role of NGOs in this is very important. Their strength lies in their manpower, informality in operations, and valuable human resources. Their ability to reach out to the community and sensitivity to local traditions is an asset in such situations.

For earthquake prediction a number of geophysical and geochemical parameters are continuously observed and some of which shows anomalously precursory changes. These include the following:

1. Land deformation, tilt, and strain are the most important changes which are observed well before the earthquakes.



2. Number of small shocks (foreshocks) increase before a main earthquake but decrease just prior to the main earthquake.
3. Velocity of longitudinal waves in the earthquake zone decreases and then becomes normal prior to an earthquake.
4. Electrical resistivity of the ground decreases.
5. Radon, a radio active gas, is found to increase prior to earthquakes.

According to the report, the initiatives of the International Fund for Agriculture Development in supporting the Self Employed Women's Association and the Government's initiative in a community based livelihood security for earthquakes and drought victims have the potential to shape the future disaster response and development projects. The author however states that the coordination between government, local NGOs, and local community initiatives for both the rescue as well as rehabilitation needs to be strengthened as this can cause delays, overlaps, and waste of relief material and efforts.

18.7.3 Cyclones:

Cyclone is a meteorological phenomenon of intense depressions forming over the open oceans and moving towards the land on the shore. In reaching the shores, it moves into the interior of the land or along the shorelines. The cyclone once formed may be active from days to weeks and affects many days to weeks and affects many areas, even countries depending on the nature and the intensity.



Fig. 4

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Globally North West Pacific regions are more prone to cyclones. The Indian Ocean is one of the six major cyclone prone regions of the world. India has a long coastline of 5700 kms, which is exposed to tropical cyclones arising in the Bay of Bengal and the Arabian Sea. The eastern coastline is more prone to cyclones as it is hit by about 80 percent of the total cyclones generated in the region. In India, cyclones originating from Bay of Bengal are more in number and intensity. In India, cyclones occur usually between April and May and also between October and December. The damage depends on the intensity of cyclone, the damage to human life, crops, settlements roads, communications, tanks, canals, and livestock. Sometimes their occurrence slow down the developmental activities of the areas.

Disaster management should aim at reducing the impact of the three main characteristics and effects of a cyclone, which are - High speed winds, Storm Surge, and Floods caused by Heavy and wide-spread rainfall. The focus therefore has to be on the following: -

- a) Understanding the mechanism of formation, development, structure, and movement of cyclones.
- b) The capability of detecting cyclones while out at sea.
- c) The capability to predict their movement and behavior.
- d) Capacity to warn vulnerable people in time.
- e) Measures for cyclone preparedness both in advance and during a cyclone.
- f) Relief and rehabilitation after the cyclone.
- g) An integrated hazard mitigation policy dove-tailed into the development plan.

The first four are essentially based on meteorology and the rest are in the field of planning, organising, and implementation.

Mitigation measures are:

- Installation of early warning systems
- Developing communication infrastructure
- Developing shelter belts



- Construction of permanent houses
- Training and educating on land use, control, and settlement planning.

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18.7.4 Landslides:

Landslides are recurring phenomena in the Himalayan region. It is a geological process which includes a wide range of mass movements, such as rock falls, deep failure of slopes, and shallow debris flows. Although gravity acting on an over steepened slope is the primary reason for a landslide, there are other contributing factors affecting the original slope stability like erosion by rivers, glaciers, or ocean waves create over steepened slopes. In recent years, however, intensive construction activity and the destabilizing forces of nature have aggravated the problem. Landslides occur as a result of changes on the slope, sudden or gradual, either in its composition, structure, hydrology, or vegetation. The changes can be due to geology, climate and weather, changing land use, and earthquakes.

1. Rock and soil slopes are weakened through saturation by snowmelt or heavy rains. Earthquakes create stresses that make weak slopes fail.
2. Volcanic eruptions produce loose ash deposits, heavy rain, and debris flows.
3. Vibrations from traffic, machinery, thunder, and blasting can trigger weak slopes.
4. Groundwater pressure acting to destabilize the slope.
5. Excess weight from accumulation of rain or snow, stockpiling of rock or ore, from waste piles, or from man-made structures may stress weak slopes to failure and other structures.

Measures to prevent landslides are drainage measures, erosion control measures such as bamboo check dams, terracing, jute and coir netting and rock fall control measures such as grass plantation, vegetated dry masonry walls, preventing deforestation, and improving forestation. A significant reduction in the hazards caused by landslides can be achieved by preventing the exposure of population and by physically controlling the land slides.



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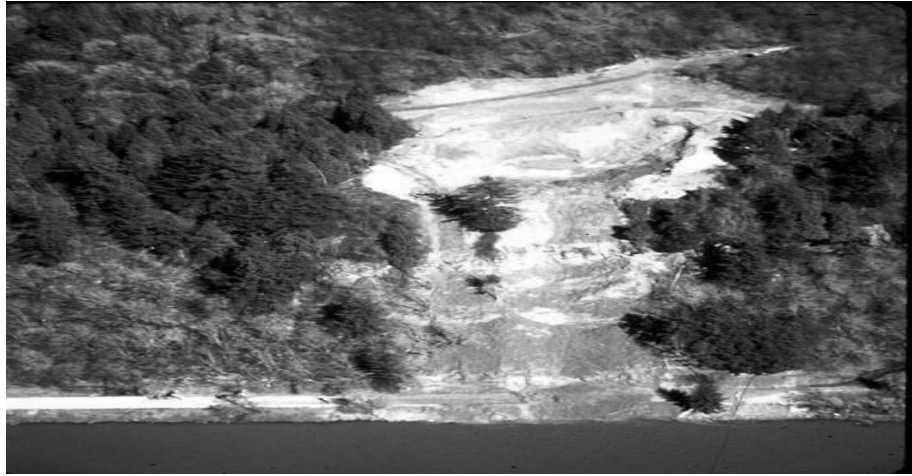


Fig. 5

18.7.5 Tsunami:

The term ‘Tsunami’ comes from the Japanese language, meaning harbor (tsu) and wave (nami). A tsunami is generated when the sea floor abruptly deforms and vertically displaces the overlying water. It is the wave disturbance that rapidly displaces a large mass of water like an undersea earthquake, volcanic eruption, or submarine land slide. Tectonic earthquakes are a particular kind of earthquakes that are associated with earth’s crustal deformation; when these earthquakes occur beneath the sea, the water above the deformed area is displaced from its equilibrium position. Waves are formed as the displaced water mass, which acts under the influence of gravity attempts to regain its equilibrium. When large areas of the sea floor elevate or subside, a tsunami can be created. Tsunami wave can travel at the speed of a commercial jet plane, over 800 km/h. They can move from one side of the Pacific Ocean to the other in less than a day. The waves can be extremely dangerous and damaging when they reach the shore. The wave travels across the ocean at speed of 500-1000 km/h. As the wave approaches the land, it compresses - sometimes up to a highest of 30ms and the sheer weight of water is enough to crush the objects in its path, often reducing the building to their foundations and scouring exposed ground to the bed rock.

Self-Instructional

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Fig. 6

One of the most severely affected areas was that closest to the epicenter, the province of Aceh on Sumatra, Indonesia. More than 130,000 people died and 36,786 were still missing in December 2005. The highest tolls were among the women and children who were in the low lying coastal areas while their husbands were at sea fishing. Over 800 kilometres of coast was severely affected, often up to five kilometres inland. At least 654 villages were damaged or destroyed, more than 500,000 people lost their homes, and more than 150,000 children were left without schools. To add to the devastation an earthquake measuring 8.7 on the Richter scale struck the west coast of Sumatra near the island of Nias on 28 March, 2005. On the morning of Sunday, 26 December, 2004, there was a severe earthquake in the Indian Ocean off the coast of northern Sumatra, Indonesia. The earthquake measured 9.0 on the Richter scale and was followed by aftershocks ranging from 6.3 to 7.0 in severity in a zone 1,000 kilometers north to the Andaman Islands. The underwater earthquake also resulted in a powerful tsunami ('soo-na-mi', from the Japanese words meaning 'harbor wave'). The wave travelled quickly under the ocean, building to a wall of water up to 10 meters high when it reached the shallow coastal waters and causing massive destruction when it hit land. Without an effective warning system and disaster plan, many people did not know to move quickly to higher ground to escape the wave and its load of debris. In some places the sea receded for hundreds of meters before the wave rushed in. Curious people looking at this strange occurrence from the

*Self-Instructional
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beaches did not recognize this as a sign of danger, and as a result many were killed by the tsunami. The tsunami caused extraordinary damage. The death toll was put at roughly 187,000, with nearly 43,000 missing and many hundreds of thousands injured and suffering. Countries lost people with the knowledge and skills that were needed for their ongoing development. Roads, bridges, water and electricity supplies, health centers and schools were destroyed.

DEALING WITH DISASTER -

- 1) Natural Disaster can strike suddenly, unexpectedly and with or without warning. It leads to temporary disruption of all social, economic and community life.
- 2) Large number of agencies like Revenue, Police, Health, Transport, Electricity, Fire brigade have a part in dealing with disaster and rehabilitation.
- 3) Effectiveness of the response depends upon how well the emergency services, both Local and Central Governments have harmonised and fine tuned their preparation and exercised their preparational arrangements - emergency procedures.

18.8 SUMMARY

India is affected by frequent and devastating disasters. Its vast expanse and varied geo climatic features add to the wide variety of disasters that strike our land. Initially disasters were considered to be interruptions in the process of development, so they were managed after occurrence in the form of relief distribution, reconstruction and rehabilitation. Pre disaster efforts for prevention, mitigation, and preparedness were almost absent in any form of government.

India has also taken great strides in disaster preparedness and risk reduction. As a partner in the worldwide movement to usher in a Culture of Prevention, India has taken strong steps to move from a relief-based response to equal emphasis on prevention, mitigation, preparedness, relief, response, reconstruction, and rehabilitation. The nodal ministry has shifted from the Ministry of Agriculture



to the Ministry of Home Affairs in 2002. The most significant change has been brought by the passage of the Disaster Management Act 2005. It has ensured the joint responsibility of both Central and State Governments for managing disasters and it provides a dedicated institutional mechanism and funding at the national, state, and district levels. The Act also provides Disaster Management Authorities at the national, state, and district levels, in which the national and state executive committees are for the implementation and there are specialized response forces at the national and state levels. India is thus going through a phase of intense preparedness for all disasters. These initiatives aim to usher in a Culture of Prevention across the spectrum of stakeholders in the short and long run.

The mission is to build a Disaster Resilient Community who will be empowered with the capacity to deal with hazards, without succumbing to its devastation and the new institutional mechanism provided by the Act aims to facilitate this process.

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18.9 SELF-ASSESSMENT QUESTIONS

1. Explain the salient features of natural disasters.
2. Describe the implications of Earthquake, Cyclone, and Tsunami disasters.
3. Explain the general threat perceptions of man-made disasters.
4. List various preparedness and mitigation measures for combating such natural disaster.

18.10 REFERENCES

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LESSON 19

PRACTICAL UNDERSTANDING OF ENVIRONMENTAL SCIENCE

NOTES

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Structure

- 19.1 Learning Objectives
- 19.2 Visit to a Local Area to Document Environmental Assets: River/Forest/Grasslands/Hill/Mountain
 - 19.2.1 Documenting Environmental Assets of each ecosystem
 - 19.2.2 River Ecosystem
 - 19.2.3 Forest
 - 19.2.4 Grassland
- 19.3 Visit to a Local Polluted Site-Urban/Rural/Industrial/Agricultural
- 19.4 Study of Common Plants, Insects, Birds
- 19.5 Study of Simple Ecosystems
 - 19.5.1 Forest
 - 19.5.2 Grassland
 - 19.5.3 Desert And Semi Arid-Areas
 - 19.5.4 Aquatic Ecosystems
- 19.6 References

19.1 LEARNING OBJECTIVES

After completing this chapter, you should be able to understand the primary concept of practicals related to basic Environment.



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19.2 VISIT TO A LOCAL AREA TO DOCUMENT ENVIRONMENTAL ASSETS: RIVER/FOREST/ GRASSLANDS/HILL/MOUNTAIN

Documenting the nature of an ecosystem gives us a deeper appreciation of its value to mankind. Each ecosystem has something different to offer us. It may contain natural resources that local people depend on; or provide important ecological functions for all of us; or have tourist or recreational potential; or simply have a strong aesthetic appeal that is difficult to quantify in economic terms. In fact it can have multiple benefits for mankind at global, national, and local levels. An ecosystem is not only used by different cultures and socio-economic groups in various ways, but has a different significance for different individuals depending on their way of life. A tribal from a wilderness setting, an agriculturalist from farmlands, a pastoralist from grasslands, or a fisherman looks on his or her environment very differently from an urban resident who is mainly focused on the management of the quality of air and water and the disposal of garbage. In many cultures, men and women will have different views and relationships with nature. For example, in rural India, it is mostly women who collect resources and see the degradation of their ecosystem as a serious threat to the existence of their family and are thus more prone to fight against processes that lead to loss of their resource base. Tribal people who live by hunting and gathering have a deep understanding of nature and what it provides for them to survive. Farmers know about utilisation of their land and water resources, and also appreciate what droughts and floods can do to their lives. A shepherd or livestock owner knows the grasslands intimately. In contrast, urban dwellers are far removed from the sites from where they get their natural resources. As products have been originated from a remote area and are collected by rural people, they cannot relate easily to the value of protecting the ecosystems from which the resources have come. In assessing an ecosystem's values, it is not enough to look at its structure and functions, but at who uses it and how the resources reach the users. One also needs to appreciate what it means to oneself. The wilderness provides a sense of wonderment for all of us, if we experience it in person. This helps to bring about a desire to conserve natural resources.



Guidelines for the study of environmental assets:

There are two parts to this study:

1. Document what you see.
2. Document the findings of what you ask local user groups.

There are several key questions that one should attempt to answer in the study of any ecosystem's natural resources:

1. What are the ecosystem's natural resource assets?
2. Who uses these and how?
3. Is the ecosystem degraded? If so how?
4. How can it be conserved?

One could go into enormous details in answering these four basic questions. You will need to refer to relevant chapters in this textbook, the guidelines provided in this Lesson, as well as field guides to plants, insects, birds, etc.

You should begin your field study by observing the abiotic and biotic aspects of the ecosystem and documenting what you see. Ask questions to local user groups about their environment. Is their utilisation sustainable or unsustainable? Look for and document signs of degradation. Finally, study the aspects that can lead to its conservation.

- Describe the ecosystem as you see it. Its structural nature, its quality and the differences one can perceive in its geographical features, and its plant and animal life. This takes time and patience. The more time one spends on a careful scrutiny, the more one begins to appreciate its intricacies.
- How does the ecosystem function? What are the linkages between different species with each other and with their habitat? Observe its food chains. Look at it as if it is an intricate machine at work.
- By interacting with local residents and multiple user groups, decide if this is sustainable or unsustainable utilisation. If it is undisturbed, why has it remained so? If it is sustainably used, how is its use controlled? If it is degraded, how did it get to this state and when? If it is seriously degraded what measures would you suggest to restore it and to what extent could it be used so that utilisation would be sustainable?

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You may not be able to observe all these questions during a single visit. Thus you will have to ask questions to local people who have a stake in the area to answer these questions. You may need the help of an ecologist, botanist, zoologist, geologist, hydrologist, or forester to get deeper insights. A historical background frequently helps to clarify many of these questions as landscapes are not static and always change over time.

19.2.1 Documenting Environmental Assets of each ecosystem

Documenting general features during the field survey: Describe the site and its features as provided in the proforma for fieldwork under the following headings – Aims and Objectives, Methodology, Observations on the site, Findings of interviews with local people, Results and Conclusions.

Documenting the special resource features of individual ecosystems

Once the general features are documented, observations pertaining to the specific features of the ecosystem must be documented. The checklist on resource use of each ecosystem can help in creating an environmental profile of an area and will help in your appreciation of the ecosystem's goods and services, which include its important assets. However, this is to be used only as a guideline and a note needs to be prepared on each finding once you have made your observations and asked local people relevant questions about the ecosystem's resources in detail. Unless one does this for several different areas, one cannot really appreciate the assets of an ecosystem in clear terms, as these are often qualitative judgements that one makes by comparing the resources available in the study area with many others.

19.2.2 River Ecosystem

Guidelines on what to look for on river resource use:

- **Observe what local people use from the river, wetland or lake:** They collect drinking water and use it for other domestic needs. They catch fish and crabs, graze their cattle, and buffaloes in or near the water. They lift water from the lake by pumps to irrigate their fields.



- **Mapping land use in terms of its water resources:** Document the pattern of land use around the aquatic ecosystem – river, tank or lake, and assess the importance of the water resources in the ecosystem. Observe that all the animals both wild and domestic must come to the water source, or have its water brought to them.
- **Field observations on a river front:**
 1. Observe a clean stretch of river in a wilderness area. The water is clear and full of life. In many pools fish dart about. Tadpoles swim around and crabs crawl along the bottom of the water.
 2. In a rural area observe all the different ways in which people use the water from the river.
 3. Observe a river in an urban area, the water cannot be used for drinking as it is dirty. Observe the water in a glass – it is coloured – can we drink it?, ‘Who has polluted it and how?’ This is a sign of unsustainable use of water.
- **Possible Observations:**
 1. Along a river in a forest observe all the different animal tracks at the edge of the water. All wildlife depends on this resource for their day-to-day survival.
 2. Identify the different fish that local fishermen have caught. Ask if the fish catch has decreased, remained the same, or has increased during the last decade or two.
 3. Resource use: Observe and document the different types of fish and other resources used by local people. Is this for consumptive or productive purposes.
 4. Observe how the ecosystem is utilised and document these assets – water distribution, fish, crustacea, reeds, plants used as food, or any other resources.
- In your report, compare and contrast an unpolluted and polluted body of water. Only the more robust species remain in polluted water while the more sensitive disappear.



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Water – The greatest of all resources:

- What do you use water for during the course of one day?
- How much do you use?
- Can you stop wasting water by using it carefully?
- How can you reduce the water you use for bathing and other uses? Discuss how wastewater can be used in the garden.
- How can water be recycled?

Observations on the site that should be recorded:

- Type – Permanent flow/seasonal flow/slow moving/rapid flow; deep/shallow.
- Qualitative aspects – Describe its abiotic and biotic aspects. Is the flow natural or disturbed by a dam upstream?
- Describe its aquatic plant and animal life.
- What are the characteristic features of its components – banks, shallow areas, deep areas, midstream areas, islands. How is the land used?

Findings on the site that should be recorded through interviews:

- What is the water used for and in what proportion? – Domestic use/ Agriculture/Industry.
- What other resources are used – fish, crustacea, reeds, sand, etc. What impact does the level of use have on the ecosystem?
- Is the water potable? If not what are the sources of pollution – domestic sewage/agricultural runoff/industrial effluents. Which of these affect it most seriously?
- Extent of pollution – Severe/high/moderate/low/nil. Explain why.
- Test the water quality. What are the results of your water quality tests?
- What efforts are made to keep the river clean, or to clean it up?
- Is its utilisation sustainable or unsustainable?



- Provide a historical profile of and changes in its environmental status by asking local people.
- Does it flood? If so how frequently? How does this affect people? What preventive steps can be taken to prevent ill effects of floods?
- How can you enhance public awareness on the need for keeping the river clean?
- How are you dependent on the river ecosystem? How is it linked to your own life?
- Results of the water analysis.

19.2.3 Forest

Guidelines on what to look for on forest resource use:

- **Assessing forest use:**

Ask local people, especially the women, what products they collect from the forest. Document what they use for household, what are sold in the local marketplace and what are taken out and sold to other areas. Fruits, leaves, roots, nuts, fuelwood, timber, grass, honey, fiber, cane, gum, resins, medicinal products, etc. are all forest products of great value.

- **Looking for signs of forest use:**

Several signs tell us how the forest is used by people. Look for human footprints and hoof marks of domestic animals, which demonstrates the dependence of man and his animals on forest vegetation. Observe the number of cattle tracks and cow dung piles, which tells us where the local people graze their domestic animals. Specially look for cattle tracks near watering places. The zigzag paths on a hillslope that have very little vegetation cover are a sign of overgrazing. People cut the branches of the trees and shrubs for fuel wood. The amount of cut stumps of branches can be used to assess the level of utilisation. If the forest is seriously lopped all around, the forest clearly appears degraded. Most of the energy required to cook meals and heat their homes in winter is forest dependent. Ask local women how far they must travel for fuelwood. Larger stumps of tree trunks show the number of poles used for building houses, or that have



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been felled and sold as timber. Observe the environment in a neighbouring village. Look for the various products used by the people or marketed by them that they get from the forest. Where do local people get their water? The presence of water in the streams is dependant on the existence of the forest.

- **Document the level of forest loss:**

Observe areas around villages where forest is overused and contrast this to the intact vegetation of Sanctuaries and National Parks. Are there signs of degradation of the canopy, formation of wasteland, or signs of soil erosion?

- **What are the products that you use in daily life that originate in forests?**

Example: water, paper, wood, medicines, etc. The oxygen we breathe is produced by vegetation only. Draw up a list of articles you use that could have been originated from a forest ecosystem. Observations on the site that should be recorded:

Identify the forest type – evergreen/semi evergreen/deciduous/dry deciduous/thorn forest.

Observations on site that should be recorded:

- Identify the forest type – evergreen/semi-evergreen/deciduous/dry deciduous/thorn forest.
- Is it a natural forest or a plantation?
- Observe its qualitative aspects – Undisturbed/partially disturbed/mildly degraded/severely degraded.
- Findings on the site that should be recorded through interviews.
- List its natural resources – goods and services. Goods - food, fuelwood, fodder, non-wood forest products, water, etc. Services - water regime, climate control, oxygen, removal of carbon dioxide, nitrogen cycle, etc.
- Who uses the ecosystem's natural resources and to what extent? List the level of use of each of its natural resources (sustainable/unsustainable). Are these used for personal use, for marketing, or for both? What proportion of the income of local people comes from the sale of fruit, fodder, wood, and non-wood forest products?



- Make a map of the study area showing the different land uses and where the resources are collected from.
- Provide a historical profile of its utilisation and changes in its environmental status by asking local people about their resource dependency.
- Is the ecosystem overused due to the number of people that depend on it, or the greed of a few, or both?
- Is it protected, if so how?
- If it is to be restored, how one can make this possible?
- What forest produce do you use in your day to day life?

19.2.4 Grassland

Guidelines on what to look for on grassland resource use:

- Utilisation pattern of the grassland: Discuss with local people how they use the grasslands, grazing cattle, cutting fodder, collecting fuelwood from the shrub cover etc.
- Grassland carrying capacity: Observe the enormous quantity of grass needed for the number of domestic herbivores dependent on it. This is an indication of the 'carrying capacity' of the grassland, ie. how many animals it can support.
- Mapping landuse in grassland areas: Near a village make a landuse map showing where the cattle are sent for grazing and for water, where people collect fuelwood, etc.
- Documenting grassland degradation: Document if there has been a change in landuse patterns during the last few decades by asking local people. Observe differences in protected and degraded areas.
- What are the products that you use that come from grasslands? Example: milk, meat, etc.

Observations on the site that should be recorded:

- Identify the type of grassland – Himalayan/Terai/semi-arid/shola/area developed for grass collection/common grazing land/forest clearing.



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- Qualitative aspects – Describe its abiotic and biotic features – Document the nature of its soil, plant and animal species (wild and domestic). How do they use their habitat?
- What changes occur seasonally?
- Findings on the site that should be recorded through interviews.
- Who uses it and to what extent?
- Estimate the extent of free grazing – cattle, sheep, goats, and their proportion.
- Extent of fodder collection.
- What is the productivity of the grassland? Estimate from local people if the fodder is not enough for their own livestock/just enough for their own livestock/enough for their own livestock and to sell to other fodder short areas.
- Provides a historical profile of its utilisation and changes in its environmental status by asking local people.
- This utilisation level sustainable or unsustainable?
- Is the grassland burned too frequently? Document why local people burn the grass.
- Can they do a rotation grazing of their common grasslands and thus manage it better?
- What products do you use from grassland ecosystems in your daily life?

19.3 VISIT TO A LOCAL POLLUTED SITE-URBAN/ RURAL/INDUSTRIAL/AGRICULTURAL

Pollution occurs from a variety of sources and affects different aspects of our environment and thus our lives and our health. Polluted sites include urban, rural, agricultural, and industrial areas.

Identify the site type and describe the sources of pollution.

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Pollution can affect:

- Air (smoke, gases)
- Water (urban sewage, industrial chemical effluents, agricultural pesticides, and fertilisers)
- Soil (chemicals, solid waste from industry, and urban areas)
- Biodiversity: effects on plant and animal life (Observations on pollution must include all the above aspects).

General observations:

The following aspects need to be observed and documented:

- The type of land or water used in the polluted area, its geographical characteristics, who uses the area, who owns it.
- Map the area to be studied.
- Identifying what is being polluted – air, water, soil; the cause (s) of pollution and the polluting agent (s).
- Assess the extent of pollution – severe/moderate/slight/nil, to air, water, or soil biodiversity.
- Assess from literature, the health aspects associated with the pollutant.
- Ask local residents about its effect on their lives.
- Make a report of the above findings

1. Solid Waste Study Site

Guidelines for the study of polluted solid waste sites:

Pollution caused due to solid waste can be seen at various places:

- A. Garbage dumps:** One of the urban or rural environmental problem sites that can be studied is a garbage-dumping area. This problem is basically due to increase in population, an over utilisation of non-biodegradable disposable consumer goods and lack of awareness of the management of waste at the household level. How much garbage is produced everyday



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is not given much thought. No one really thinks about where the garbage goes or what happens to all the things we throw away.

Garbage is a source of various diseases. Improper handling of organic waste leads to a large population of flies, cockroaches, and rats that are responsible for spreading diseases. Products like plastics are not degraded in nature and hence remains for a long time in the environment, thus adding to the need for more dumps. For many years waste has also been dumped into oceans, rivers, or on land. These methods of disposing off waste contribute to contamination of soil, groundwater under the dumping site, foul up the air, and aid the spread of diseases.

- B. Households:** The garbage generated in our homes is termed domestic waste, while a community's waste is referred to as municipal waste. This is classified as kitchen waste, which is degradable, wet waste and non-biodegradable recyclable home waste which consists of plastic, glass and metal. Observe and document what happens in homes of different economic groups. What happens to your own household waste? Could it be managed better?
- C. Agriculture:** Agricultural waste consists of biomass including farm residues such as rice husk, straw, bagasse, etc. This biomass could be effectively used for generating power or producing paper. Waste material from fields includes fertilizers and pesticides that are a serious health hazard.
- D. Industries:** Industrial solid waste includes material from various industries or mines. Industries produce solid wastes during manufacturing processes. Some of these are chemicals that have serious environmental ill effects, as they are toxic. Visit an industry and ask what are the waste products and how they are disposed off. The waste generated during mining is non-biodegradable, it remains in the environment nearly indefinitely. Solid waste is also generated as a result of excavation and construction works.
- E. Hospitals:** The waste generated from hospitals contains cotton dressing and bandages with blood or other tissue fluids and pus, all of which can contain pathogens. It can spread bacteria, fungi, and viruses. Used needles, syringes, bottles, plastic bags, operation theatre waste, such as



tissues, blood, plastic disposable equipment, etc. all needs very careful disposal. The hospital should have a waste separation system at source into biomedical waste, glass, plastics, etc. The biomedical waste can be autoclaved or incinerated so that microorganisms are killed.

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2. Water Pollution Site

Guidelines for the study of polluted water sites:

- Observe if the river/lake/tank can be considered unpolluted/slightly polluted/moderately polluted or severely polluted by looking at the water and by simple tests using a water monitoring kit.
- Document the name of the river and the nearby urban or industrial site from which the pollution is generated.
- Is there any urban garbage dumped on the bank?
- Are there industrial units near the site?
- Do the industries discharge their waste water into the site? Is this treated or untreated?
- What is its colour and odour?
- Are there any sources of water contamination from the surface runoff or from adjacent agricultural land on which fertilizers and pesticides are used?
- Ask fishermen if this has affected their income.
- Identify plants, birds, and insects found on the banks.

3. Air Pollution Site

Guidelines for the study of polluted air sites:

Air pollution sites include cities due to traffic congestion in urban centers and industrial areas due to gaseous products released during manufacturing processes

- Ask people from the area the effects on their lives.
- How can this be reduced?
- How can you make more people aware of this issue and the effects on their health.



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19.4 STUDY OF COMMON PLANTS, INSECTS, BIRDS

Guidelines for the study:

These taxa have been selected as they occur nearly everywhere whereas one may have to visit a National Park or Sanctuary to see mammals or reptiles. Several plants, insects, and birds can be seen around an urban or rural setting where there is some vegetation. If you have an opportunity to visit a National Park or Sanctuary, you can add other animals.

One needs a little equipment – a journal to take notes, preferably a pair of binoculars, field guides to identify plants, insects, birds, reptiles, and mammals. These are available from Bombay Natural History Society (BNHS).

Plants:

- Identify and list common plant species at the study site (at least 20; 10 trees, 5 shrubs, 5 herbs).
- Identify if there are rare species by using a field guide or asking a botanist.
- Identify and list the types of plants – trees/shrubs/climbers/ground cover – herbs, grasses. Observe their abundance levels.
- Describe five plant species. Document the characteristic features that help in identification of the selected species:

Specific characteristics of leaves/flowers/fruit/seeds.

- Describe the plant's role in the ecosystem.
- How is it used and by whom?
- Is it being collected sustainably or over harvested?
- Is it common or rare?
- If rare, why?
- Is it a keystone species? If so why?



Animals:

- Identify and list all the species you see in the study site.
- What are the major field identification features of the common animals and birds that you observe?
- Look for and document for each group; insects, birds (mammals if possible) at least 10 species.
- Document the characteristic features for each of the ten species and record the following:
 - a) The role of the species in the ecosystem: What role does the species play in nature – producer, herbivore, carnivore, decomposer, pollinator, seed dispersal agent, pest, etc.
 - b) The level of abundance at the site – Classify as abundant/common/rare/very rare.
- Watch and document the area unobtrusively to observe all the linkages between the different species and between a species and its habitat. What role does each species play in the food chain and energy pyramid?
- Observe the habits of each of the selected species such as feeding behaviour, nesting (for birds), breeding, territorial behaviour, etc.
- Refer to a relevant field guide and document the following:
 - a) The distribution of each of the selected common species in the country.
 - b) The current status from a field guide – abundant, common, uncommon, rare, endangered. If rare, is it on the endangered list?
- Is it used by people? For what purpose?
- How can it be protected

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19.5 STUDY OF SIMPLE ECOSYSTEMS

General guidelines on aspects that can be observed and documented during ecosystem field studies:

1. The major questions that must be addressed during a field visit to any ecosystem such as a forest, grassland, semi-arid, desert, hills, mountain ranges, lake, river, or seacoast include:
 - What is the ecosystem called on the basis of its typical features? What are its abiotic and biotic characteristics?
 - Are its goods and services being misused or overused? What are the signs that can be observed of degradation of the ecosystem that have occurred in the area? Deforestation, pollution of a waterbody, soil erosion, etc. are signs of degraded ecosystems.
 - How can this degradation process be prevented by sustainable use of the ecosystem's goods and services by changing one's own habits, such as by saving water, electricity, paper etc?
 - How can we all care for our 'mother earth' in our own way? Many small actions together reduce the adverse impacts of human activities on the ecosystems.
2. **Observing the Water Cycle:** During a monsoon field trip observe the effect of the rain.
 - The type of vegetation is an indicator of the amount of rainfall. Classify the ecosystem: forest/grassland/semiarid/desert type on the basis of rainfall.
 - Observe how rain percolates into the subsoil. This recharges ground water, which charges wells, streams, and rivers.
 - Document if the rain is eroding the soil. This can be judged by observing if the water is brown in colour. The colour is an indicator of the extent of soil erosion and is darker wherever plant cover has been destroyed. It takes thousands of years for new soil to form. Excessive silt eventually changes the course of the river and leads to flooding of surrounding land.



- 3. Observing the Carbon Cycle:** Since plants take up carbon dioxide, which we exhale, and split it into carbon and oxygen, which we breathe, we are dependent on the plant life on earth. Eventually large-scale deforestation could make life on earth impossible. Document this as an ecosystem service.

Carbon is a component of the food we eat in the form of carbohydrates, which come from plant material. Thus we need plants to give us oxygen and food, without which we cannot survive.

- 4. Observing the Oxygen cycle:** While on the field trip focus attention on the amount of green material that plants contain. Without this there would not be enough oxygen for animals to breathe. Sunlight is essential for plant photosynthesis, which produces new leaves, branches and the growth of the trunks of trees. It leads to growth of grass and herbs every year.

Sunlight is essential for plant growth in the water, including microscopic algae and underwater vegetation which is the food producer for all aquatic forms of animal life.

- 5. Observing the Nitrogen Cycle:** Observe the quantity of dried leaves on the ground in a forest, or the dried leaves of plants planted in the area that have collected as detritus. This material can be seen to be decaying. Ants, beetles and worms that feed on this dead material are breaking it up into small fragments. Microscopic bacteria and fungi are acting on this material to convert it into nutrients for plants to grow.

- 6. Observing the Energy Cycle:** Look for the different types of insects and birds in the trees. Frugivorous birds feed on fruit, insectivorous birds and spiders feed on insects. These form food chains. There are thousands of such food chains in an ecosystem. These inter-linked chains can be depicted in the form of a 'web of life'. Observe that in our surroundings there is a great amount of plant material. There is much less animal life in which there are a relatively larger number of herbivores than carnivores, which live on herbivorous animals. Estimate and document the differences in the number of plants, herbivores and carnivores in an area. This can be depicted as a food pyramid.



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19.5.1 Forest

Field Visit to a Forest – Depicting the nature of the ecosystem:

Visit the nearest or most convenient Reserved Forest, National Park, or Wildlife Sanctuary. Meet the Forest Official to explain your study. Check if there is an Interpretation Centre where there may be local information. Ask for brochures or other material. Officials may agree to address a group of students. Observe the forest type. Make notes on the ecosystem.

- **Classify the forest type:** During the field visit to the forest identify which type of forest is found in the area. Is there only one type or are there several types? If so why? Coniferous, deciduous, evergreen, thorn forest, and mangrove are some examples.
- **Interpreting the connection between abiotic and biotic aspects of the ecosystem:** Observe differences in vegetation types during the field visit and relate this to abiotic features such as temperature, rainfall, soil, and topographic patterns wherever possible.
- **Understanding food chains and food pyramids:**
 - Observe the abundance of different species in the ecosystem. Observe which plants are found commonly in the forest. Only a few species are very abundant but there are a large number of less common species of trees, shrubs and climbers and small ground plants that add to the diversity of plant life in any forest.
 - Observe and document the names of animals seen. Classify them as mammals, birds, reptiles, amphibian, or insects. Classify these into herbivores and carnivores. If these are counted, you will appreciate that there is a relative abundance of herbivores over carnivores.
- **Identify the structural levels in a forest:** Identify the layers of the forest. Draw profiles of the structure and label the levels.
Ground–trunk–branches–canopy.
- **Document the micro-habitat for species in different levels of the forest:** Observe which animal uses different parts of a forest habitat. Some live

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on the ground among the fallen leaves (worms and insects such as ants, termites and beetles), others live in the middle layer on branches and tree trunks (lizards and woodpeckers), many others live in the canopy of the tree tops, (such as fruit and nectar dependant birds such as sunbirds, parakeets and mynas. There are insectivorous birds, (flycatchers, drongos and bee eaters) in the canopy. Several insects live under the ground. If one turns over dead leaves on the forest floor there are a large number of animals (millipedes, ants, beetles, etc.).

Document what you have seen and estimate their abundance at different levels.

- **Observe food chains and interpret the food web.**

Field Observation – examples of food chains that are easily seen:

Flower→butterflies→spiders

Flower→sunbirds→birds of prey

Fruit→parakeet→birds of prey

Seeds→rodents→birds of prey

Flowers→bees→bee eaters

Seeds→munias→small carnivorous mammals and birds of prey

Leaves→monkey→leopard

Grass→chital→tiger

Observe what all the animals are feeding on and reconstruct as many food chains as possible.

Observe that a single species can play a role in several food chains. Thus the chains form a food web.

Write about what you have seen about the food chains and food web in the area.

19.5.2 Grassland

- **A field visit to a grassland:** Observe the variety of plant and animal life in the grassland. Document the food used by each animal that is identified.



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- **Describe the seasonal changes in the grassland:** Describe how the grassland would look in different seasons.
- **Describe the anticipated changes in colour and the condition of grasses:** Growing phase, flowering phase, dyeing phase, or dry phase.
- **Abundance of grassland species:** Try to count the number of grasshoppers that jump out of a disturbed 1sq.m quadrant on the ground. Count at least 20 such quadrants. You may find this hard! Repeat the count for ants, beetles, etc. This will be nearly impossible, as there are just too many of these insects. This will demonstrate the great abundance of insects in the grassland.

Compare this to the much smaller number of first order consumers - birds and mammals that can be counted in the grassland. The predators, mammals, and birds of prey— raptors, are least abundant.

- **Birdwatching in grasslands:** Make a checklist of common grassland birds by identifying them from the Book of Indian Birds, by Salim Ali. Read what each species feeds on.
- **Observing the insect world:** Observe how the ants live and collect food. Observe how beetles behave in the grassland. Observe the abundance of grasshoppers, beetles, and ants. Compare this with the number of their predators.
- **Observe a spider catch its prey:** See the different types of webs. Tunnel web spiders make a tunnel and sit inside waiting for prey, which are pulled in and eaten. Other spiders in the grassland make small orbwebs that have radial and spiral threads. Some spiders build a colony which is like a mass of web material.
- **Document animal behaviour:** Make a general list of behavioural patterns of all the animals, birds, and insects you see. What are they doing? How and where did you find them in the grass? What is their relationship to the grassland as a habitat?
- **Understanding grassland food chains:** Identify as many plant and animal species. Use the list to form as many food chains as possible.



19.5.3 Desert And Semi Arid-Areas

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- **Observe desert and semi-arid landscapes:** Observe the sparse but specialised nature of vegetation in a desert or semi-arid landscape. Document the number of animal species that are seen in the vicinity. There are very few compared with other types of ecosystems.
- **Observe the fauna of semi arid country:** Identify the birds and insects which are most easily seen. Document how each species is using its habitat. What do these species feed on in this harsh environment?
- Observe typical species such as dung beetles that roll dung into a ball in which they lay their eggs so that their young ones get food.
- Observe birds of prey that use this ecosystem.
- There are rare birds in a few areas such as the Great Indian Bustard.
- There are rare mammals such as the wolf.

19.5.4 Aquatic Ecosystems

Document the nature of aquatic ecosystems: Visit an aquatic ecosystem such as a pond, lake, river or seacoast. Observe if the water is clean or polluted. A simple kit can be used to study the water quality.

1. **Studies on the ecology of a pond:** Make observations on a seasonally active pond if possible on several occasions before, during, and after the monsoon. Document the seasonal changes in the plant and animal life.
 - **Observe the vegetation zones at the waters edge:** Observe the different zones of vegetation - grasses on the periphery, emergent reeds, floating vegetation, and underwater plants in the pond.
 - **Seasonal field observations on a pond:** Early stage - soon after the pond fills with water in the monsoon: Observe algae and microscopic animals. These can be observed under the microscope.
 - **Fully active phase:** Submerged and emergent vegetation – fish, frogs, snails, worms and aquatic insects.



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- **Shrinking phase:** Drying aquatic plant life with dead and dying plant material and terrestrial plants growing on the exposed mud of the pond.
 - **Dry phase:** Overgrown with grasses and shrubs with hidden dormant animals in the mud, which cannot be seen.
 - **Laboratory exploration:** Observe water from the pond in a glass. Document its colour and what it contains. Observe water from a pond under the microscope. There are a large number of algae and zooplankton that form the basic food chains of the aquatic ecosystem.
2. **Observations on a Lake ecosystem:** Document the way in which different water birds use the various habitats both on the shore and in the water. Each of the different species of aquatic birds shares its habitat with only a few other species. Each species specialises in certain types of food and feeds at different depths. The length of the legs of different wading birds is an indicator of the depth at which they feed. The length of their beaks indicates the depth of mud or sand into which they can probe.
- **Diversity and abundance of life:** Make a checklist of all the visible aquatic flora and fauna. Identify those that are most abundant. Observe and document the food chains. Estimate or count the population (abundance) of different species observed in the aquatic ecosystem.
3. **Observations at a wetland:** Visit a wetland. Observe the varied vegetation zones within the ecosystem. Document and map its vegetation patterns – Underwater/emergent/floating/none. Describe if the water is clean or turbid. Describe the level of algal growth and weeds. What is the nature of its bed – rocky/sand/silt/mud/mixed (in what proportion?). Develop a map of the aquatic ecosystem vegetation and its relationship to species of aquatic birds. Ask local fishermen to show you their fish catch. Observe the ducks, waders, and other birds. These are most abundant in the winter as most of them are migrants from across the Himalayas.
4. **Observation on a field visit to a beach:** Beaches can be sandy, rocky, shell-covered, or muddy. On each of these different types, there are several specific species, which have evolved to occupy a separate niche.
- Observe all the different crustacea such as crabs that make holes in the sand.



Observe how the various shore birds feed on their prey by probing into the sand.

- 5. Observations at a river:** Depending on the location of the river, the study can demonstrate its ecological status.

The river is a dynamic system with seasonal fluctuations in flow rates that affect its plant and animal aquatic life as well as along its banks. Observe and document how life is dependent on the river's integrity

- 6. Hill/Mountains:** The ecosystem of the hill you are observing is linked to its altitude, slope, soil characteristics, vegetation, and animal life. It has different vegetation patterns that create specific microhabitats for a variety of fauna. The habitat changes seasonally. What do you expect will occur in three months, six months, and nine months from the present scenario?

Example: Guidelines for a write-up on a HILL/MOUNTAIN ecosystem

- Describe the hill – slope, soil, watercourses, etc.
- Describe its various plants and animals.
- Observe and document its food chains.
- Describe the water cycle, the nitrogen cycle, the energy cycle, and the detritus cycle with specific reference to the hill/mountain ecosystem.
- What would happen if all the domestic animals were to be prevented from grazing?
- What would happen if ants were to be eliminated from this ecosystem?
- What would happen if all the vegetation is removed from the slopes?

19.6 REFERENCES

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